

***Logic Manager
Parameter Reference
Dictionary***

LM09-640

Implementation
Logic Manager

***Logic Manager
Parameter Reference
Dictionary***

LM09-640
R680
07/07

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About This Publication

This publication defines the parameters for the Logic Manager (LM) data point types in the **TotalPlant** Solution (TPS) system. Additional information about the use of these parameters in data acquisition and control is in *LM Control Functions*.

This publication supports **TotalPlant** Solution (TPS) system network Release 680. TPS is the evolution of TDC 3000^X.

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1. INTRODUCTION

1.1 PURPOSE

This publication defines the user-visible parameters that exist in the **TotalPlant** Solution (TPS) Logic Manager (LM). It also provides listings of parameters that are applicable to various LM point types.

For information on how the parameters are used in LM data points, refer to the *Logic Manager Control Functions* manual in the *Implementation/Logic Manager* binder.

1.2 USE OF THIS PUBLICATION

Use this publication as you build data points, and during operation when detailed information about LM parameters is required.

For use in data point configuration, this publication provides definitions for each entry that can be made on the *Logic Manager Forms* in the *Implementation/Logic Manager* binder, and in the Parameter Entry Displays at the Universal Station.

For use in process operation, this publication provides information about the parameters that appear for the process data points and LM node-specific data points on the displays of Universal Stations that are running with the Operator personality.

1.3 PARAMETER DEFINITION FORMAT

In this dictionary, the parameter definitions are listed in alphabetical order according to the parameter name, which can be up to eight characters in length. Each parameter in this publication is defined using the format shown below for the **ALMOPT** parameter, as an example. The following paragraphs describe the entries that appear within each parameter definition.

ALMOPT (DigComp)

Type: **E:\$ALMOPT** **Alarm Option**—Defines the types of alarms that can be detected by this data point.
Lock: **Eng/PB** Refer to the *LM Control Functions* manual for a detailed description.
Default: **None**
PtRes: **LMM**
Range: **None** (No alarm detection)
Offnorml (Off Normal; alarm if PV and PVNORMAL are not the same state)
Cmddis (Command Disagree; alarm if field device did not respond to commanded state within configured feedback time; also detects uncommanded changes in the field device)

Helpful Hint:	1. Offnorml alarm can be configured only if the point has at least one input. 2. Cmddis (and uncommanded change) alarm can be configured only if the point has at least one input and one output.
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For many parameters, the function of the parameter is described using the long name of the parameter name (**Alarm Option**), followed by a description as shown in the above example. Some parameters in this dictionary do not have functional descriptions following the long name; this is because the long name of the parameter sufficiently describes the parameter function.

Type

This entry is the data type that defines how the parameter is viewed by the system. The following data types are used in this dictionary:

- **E:.....**—Enumeration; the value for the parameter is chosen from a set of predefined character strings. In the above example, the enumerations of **\$ALMOPT** are **None**, **Offnorml**, and **Cmddis**.
- **Ent.Prm**—consists of a 1-8 or 1-16 character tag name (depending on the option selected), a period, and a 1-8 character parameter name.
- **Integer**—a 16-bit whole number that does not contain a decimal point.
- **Logical**—a binary type with the values of ON (True) and OFF (False), or 0 (Off) and 1 (On).
- **NaN**—although not a data type, is used to represent "Not A Number" and is stored in IEEE format.
- **Prm_Id**—1-8 character parameter name.

- **Real**—a 32-bit floating-point number in IEEE format.
- **String_L**—a character string of maximum length = L. Same as Ascii_L.
- **Universal Ent.Prm**—Universal Entity Parameter Identifier. It is basically the same as **Ent.Prm**, but the entity name can be entered as an external 16-character tag name or as a UCN node's internal hardware reference address. The hardware reference address syntax can be used to access point parameters (within this same LM) that are untagged or tagged.

Lock

The access lock defines "who" or "what" can change the parameter's value or option and the access level defines "who" or "what" is requesting a parameter value or option change. For example, if a requestor with an access level of Supr tries to change a parameter that has an access lock of Engr, the request will be denied. The two charts below describe how access levels and access locks work.

<i>Access Level</i>	<i>Used by Who or What When a Parameter Change Request is Made</i>					
Oper	Operator					
Supr or Sup	Supervisor					
Engr, Eng, or Eg	Engineer					
Cont	Continuous_Control (from a Module on the LCN)					
OnProc	On Process					
Prog	User-written programs in an LCN node					
PtBld or PB	Point_Builder (Data Entity Builder)					
<i>Access Lock</i>	<i>Access Level of Requestors That can Change the Parameter</i>					
Oper	Oper	Supr	Engr	Cont	Prog	PtBld
Supr		Supr	Engr	Cont	Prog	PtBld
Engr			Engr	Cont	Prog	PtBld
OnProc	Oper	Supr	Engr			
Sup/Eg		Supr	Engr			
EgOnly			Engr			
Prog				Cont	Prog	PtBld
Eng/PB			Engr			PtBld
PtBld						PtBld
View (Read Only)						

Default

The default for the parameter is the default value assigned by the system. The system automatically enters the default value for a parameter when a range or a selection is not entered for a parameter during point building. The default values are also shown on the configuration forms and parameter entry displays.

PtRes

This defines where the parameter physically resides. The following residency locations are used in the parameter definitions:

<i>PtRes</i>	<i>Definition</i>
NIM	Network Interface Module
LMM	Logic Manager Module

Range

This defines the range of the value that can be entered for this parameter. The integers that precede enumeration values are sometimes needed by advanced CL users. These integers specify the member's position within the set (i.e., the ordinal).

Helpful Hint

Some parameter definitions contain a *Helpful Hint* box at the end of the definition. This box contains additional information about the parameter, such as prerequisites, etc.

1.4 PARAMETERS PER POINT TYPE

In addition to the parameter definitions, this dictionary also contains listings of the parameters that are applicable to each LM point type, and are defined in Section 2.

1.5 FULL POINTS AND COMPONENT POINTS

Separate functional elements of the LM are used to implement various parts of control strategies. Each of these functional elements can be assigned a user-defined tag name to allow for location-independent reference to the data associated with that function. For example, point tags are assigned by the user for analog input and analog output slots.

1.6 ABBREVIATIONS

The LM provides a configurable parameter called PNTFORM (Point Form) that allows the user to define which points are to be used as the primary operator interface for point data. The PNTFORM parameter provides the user with two choices for point form: "Full" and "Component." Points that are configured as having "Full" point form include descriptor data and alarm-related parameters. This information is needed when a point is to be used as the primary operator interface to the point's data.

Points that are configured as having "Component" point form do not require descriptor data and alarm-related parameters, because this type of information is suppressed for "Component" points. This point form should be used for points that provide inputs to the "Full" point and also for those points that handle the outputs from the "Full" points. "Component" points should be used as part of a "Full" point that has been designated a primary operator interface point.

AnalgIn	Analog Input Data Point
AnalgOut	Analog Output Data Point
AO	Analog Output
AutoMan	Auto Manual Station algorithm
DI	Digital Input
DigComp	Digital Composite Data Point
DigIn	Digital Input Data Point
DigOut	Digital Output Data Point
DO	Digital Output
Flag	Flag Data Point
Logic	Logic Data Point (Slot)
LCN	Local Control Network
LM	Logic Manager
LMM	Logic Manager Module
NIM	Network Interface Module
Node Specif.	Node Specific
Timer	Timer Data Point
UCN	Universal Control Network
UCN Node	Universal Control Network Node

1.7 CL ACCESS

1.7.1 CL Restricted Parameters

The following parameters are not accessible to PM, APM, and HPM CL sequences. They are not *directly* available to AM/CL sequences. Access to AM/CL is through custom data segment parameters attached to AM regulatory points as described below.

NODESTS
NODETYP
UCNRECHN

These parameters are available to user schematics using the NIM reserved data point. For example, \$NMuuBnn.param, where uu = UCN number and nn = UCN node number.

AM/CL programs can access the restricted parameters as Regulatory Point General inputs (using ordinary point parameter access). They must be transferred to parameters of AM regulatory points.

For Enumerations (NODESTS, NODETYP, and UNRECHN,) a custom data segment is created to allow the parameters to be referenced as general inputs and transferred to user-defined parameters (of a RegCtl Point) that can be accessed by CL.

1.8 CHANGING NON-CONTROL PARAMETERS ON-PROCESS

With R640, the user can change non-control parameter values of an LM point without making a point inactive by using one of the following tools:

- CL/AM
- Custom schematic or DATACHNG schematic
- Alter Parameters command from the Command display
- Standard Detail display

This will save time and avoid an unnecessary shutdown of an operating loop.

However, if you are using the DEB to change the value, it will still be necessary to make a point inactive.

The non-control parameters are listed below in alphabetical order.

EUDESC*	MSGTXT*
GENDESC*	OVERVAL
KEYWORD*	PTDESC*
LOGICSRC*	STATETXT*

Indicates that the parameter cannot be changed from a standard display (Detail Display, Group Display, or Change Zone). The parameters that do not have the asterisk () can be modified from the Detail Display and their values will be updated on the standard display as they are modified.

CL/AM

CL/AM can change the value of the parameters while the point is active.

The STATETXT parameter cannot be changed using CL/AM because the access lock of the STATETXT parameter is Engr/PtBld.

The following example shows how CL/AM can change parameter values for an LM point, an MC point, and an AM point:

```
--Start of CL/AM file
--Modifying the value of KEYWORD of an LM point,
--LBOXCLR of an MC point
--and OVERVAL of an AM point.
BLOCK CHNGPAR (point AMCUS1; at general)
EXTERNAL LM27DC4
EXTERNAL MC_R001
EXTERNAL AMREG1
SET LM27DC4.KEYWORD = "DEVNTL4"
SET MC_R001.LBOXCLR = BLUE
SET AMREG1.OVERVAL=21
END CHNGPAR
--End of CL/AM file
```

Displays

When the value of a non-control parameter has been changed, the standard display showing the parameter value should be called up again to see the parameter value changes that are made by a schematic or by the Alter parameters command.

Alarming

Alarms generated prior to the parameter change will display the previous value. Alarms generated after the parameter change will display the new value.

Journals

When the parameter value is changed by the Alter parameters command from the Command Display, the change is journaled in the Engineering Change Journal.

When the parameter value is changed by a schematic, the value change is journaled in the Process Change Journal.

2. PARAMETERS PER POINT TYPE

This section contains listings of parameters that are applicable to each data point type in the LM.

2.1 ANALOG INPUT (AnalogIn)

The parameters of the Analog Input Data point are listed below in alphabetical order. (F) indicates that the parameter is applicable when the PNTFORM = Full.

\$AUXUNIT(F)	PV PVALDB(F)
\$ALDLOPT(F)	PVALDBEU(F)
\$CDETAIL	PVAUTO
\$CHKPNT	PVAUTOST
\$PTRESID	PVCALC
\$PVSTREC	PVCHAR
\$REALARM(F)	PVCLAMP
ALENBST(F)	PVEUHI
ASSOCDSP	PVEULO
BADPVFL	PVEXEUHI
BADPVPR(F)	PVEXEULO
CCSRC(F)	PVEXHIFL
CONTCUT(F)	PVEXLOFL
DISPTYPE	PVFLTPT
DLYTIME(F)	PVFORMAT
EUDESC	PVHIFL
HIGHAL(F)	PVHIPR(F)
HIGHALPR(F)	PVHITP(F)
INPTDIR	PVLOFL
KEYWORD	PVLOPR(F)
LASTPV	PVLOTP(F)
LOCUTOFF	PVP
NODENUM	PVRAW
NODETYP	PVRAWHI
NTWKNUM	PVRAWLO
OVERVAL(F)	PVSOURCE(F)
PLCADDR	PVSRCOPT(F)
PNTFORM	PVSTS
PNTSTATE	PVTEMP
PNTTYPE	PVTV(F)
PRIMMOD(F)	PVTVP(F)
PTDESC	SLOTNUM
PTEXCST	TF
PTINAL	UNITNUM

2.2 ANALOG OUTPUT (AnalgOut)

The parameters of the Analog Output Data point are listed below in alphabetical order. (F) indicates that the parameter is applicable when the PNTFORM = Full.

\$CDETAIL	OP OPFINHI
\$CHKPNT	OPFINLO
\$MODESEL	OPFINAL
\$OPTOL	OPTDIR
\$PTRESID	OUTIND
ASSOCDSP	PLCADDR
CASREQ(F)	PNTFORM
DISPTYPE	PNTSTATE
EUDESC	PNTTYPE
KEYWORD	PRIMMOD
MODATTR(F)	PTDESC
MODE(F)	PTEXECST
MODEAPPL(F)	PVFLTPT
MODEPERM(F)	RCASOPT(F)
NMODATTR(F)	REDTAG(F)
NMODE(F)	RINITREQ(F)
NODENUM	SLOTNUM
NODETYP	UNITNUM
NTWKNUM	

2.3 NODE SPECIFIC (Node Specif.)

The parameters of the Logic Manager Node Specific Data Point are listed below in alphabetical order.

\$COMCKPT	NN NNLSBA
\$LADDER	NNUMERIC
\$RECOVER	NOAINPTS
\$UCNANSY	NOCINPTS
\$UCNASIL	NOCOPTS
\$UCNATSP	NODEASSN
\$UCNBNSY	NODENUM
BADPVTXT	NODESTS
CHIPNOPR	NODETYP
CRPPXORN	NODFSTAT
CRUCNORN	NTIMER
FL	PMMCMD
FLLSBA	PMMCMDTL
LMNAME	PMMFLPOS
LSPPXORN	PMMOPER
LSUCNORN	PMMSFST
MOVPVTXT	PMMSTS
NAISLOT	PVFLTPT
NAOSLOT	SCANRATE
NDCSLOT	UCNRECHN
NFLAG	
NLOGSLOT	

2.4 FLAG

The parameters of the Flag Data Point are listed below in alphabetical order. (F) indicates that the parameter is applicable when the PNTFORM = Full.

\$AUXUNIT(F)	NTWKNUM
\$CDETAIL	OFFNRMMPR(F)
\$CHKPNT	PLCADDR
\$PTRESID	PNTFORM
\$REALARM(F)	PNTSTATE
ALENBST(F)	PNTTYPE
ALPRIOR(F)	PRIMMOD
ASSOCDSP	PTDESC(F)
BOXCLR	PV PVFL
CCSRC(F)	S0BOXCLR
CONTCUT(F)	S1BOXCLR
DISPTYPE	SLOTNUM
EIPPCODE(F)	STATE0-1
EUDESC(F)	STATETXT
HIGHAL(F)	UNITNUM
HIGHALPR(F)	
KEYWORD(F)	
NODENUM	
NODETYP	

2.5 NUMERIC

The parameters of the Numeric Data Point are listed below in alphabetical order. (F) indicates that the parameter is applicable when the PNTFORM = Full.

\$CDETAIL
\$CHKPNT
\$PTRESID
ASSOCDSP
DISPTYPE
EUDESC(F)
KEYWORD(F)
NTWKNUM
NODENUM
NODETYP
PLCADDR
PNTFORM
PNTSTATE
PNTTYPE
PRIMMOD
PTDESC(F)
PV
PVFORMAT
SLOTNUM
UNITNUM

2.6 TIMER

The parameters of the Box Timer Data Point are listed below in alphabetical order. (F) indicates that the parameter is applicable when the PNTFORM = Full.

\$CDETAIL	PNTSTATE
\$CHKPNT	PNTTYPE
\$PTRESID	PTDESC
ASSOCDSP	PTEXECST
COMMAND	PV
DISPTYPE	PVFORMAT
EUDESC	PVSLTSRC
KEYWORD	RV
NODENUM	SLOTNUM
NODETYP	SO
NTWKNUM	SP
PLCADDR	SPSLTSRC
PNTFORM	STATE
	UNITNUM

2.7 DIGITAL COMPOSITE (DigComp)

The parameters of the Digital Composite Data Point are listed below in alphabetical order. (F) indicates that the parameter is applicable when the PNTFORM = Full.

\$AUXUNIT(F)	HIGHAL(F)	OPSTTEXT
\$CDETAIL	HIGHALPR(F)	OROPT
\$CHKPNT	I0-2	P0-2
\$MODESEL	INITMAN	PISRC
\$REALARM(F)	KEYWORD(F)	PNTFORM
ALENBST(F)	LMSRC	PNTSTATE
ALMOPT(F)	LOCALMAN	PNTTYPE
ALPRIOR(F)	MODATTR	PRIMMOD(F)
ASSOCDSP	MODE	PTDESC
BADCTLFL	MODEAPPL	\$PTRESID
BADCTLPR(F)	MODEPERM	PTEXECST
BADPVFL	MOMSTATE	PTINAL
BADPVPR(F)	MOVPVFL	PV
BADPVTXT	NMODATTR	PVAUTO
BOXCLR	NMODE	PVFL
BYPASS	NODENUM	PVNORMAL(F)
CCSRC(F)	NODETYP	PVSOURCE(F)
CMDDISFL	NODINPTS	PVSRCOPT(F)
CONTCUT(F)	NODOPTS	PVSTATES
D1-2	NONECONF	REDTAG(F)
D1_0	NONE_OP1	S0BOXCLR
D1_1	NONE_OP2	S1BOXCLR
D2D1_00	NONE_OP3	S2BOXCLR
D2D1_01	NONETXT	SLOTNUM
D2D1_10	NOSTATES	SO
D2D1_11	NTWKNUM	ST0_OP1-3
DISRC	OFFNRMFL	ST1_OP1-3
DISPTYPE	OFFNRMPR(F)	ST2_OP1-3
DODSTN	OISRC	STATE0-2
EIPPCODE(F)	OP	STATETXT0-2
EUDESC	OPFINAL	UNCMDFL
EVTOPT(F)	OPRATRFL	UNITNUM
FBTIME(F)		
FINSTTXT		

2.8 DIGITAL INPUT (DigIn)

The parameters of the Digital Input Data point are listed below in alphabetical order. (F) indicates that the parameter is applicable when the PNTFORM = Full.

\$AUXUNIT(F)	NTWKNUM
\$CDETAIL	OFFNRMFL(F)
\$CHKPNT	OFFNRMPR(F)
\$PTRESID	PLCADDR
\$REALARM(F)	PNTFORM
ALENBST(F)	PNTSTATE
ALMOPT(F)	PNTTYPE
ALPRIOR(F)	PRIMMOD(F)
ASSOCDSP	PTDESC
BADPVFL	PTEXECST
BOXCLR	PV
CCSRC(F)	PVAUTO
CONTCUT(F)	PVFL
DISPTYPE	PVNORMAL(F)
DITYPE	PVNORMFL(F)
DLYTIME(F)	PVRAW
EIPPCODE(F)	PVSOURCE(F)
EVTOPT(F)	PVSRCOPT(F)
EUDESC	S0BOXCLR
HIGHAL(F)	S1BOXCLR
HIGHALPR(F)	SLOTNUM
INPTDIR(F)	STATE0-1
KEYWORD	STATETXT
NODENUM	UNITNUM
NODETYP	

2.9 DIGITAL OUTPUT (DigOut)

The parameters of the Digital Output Data point are listed below in alphabetical order. (F) indicates that the parameter is applicable when the PNTFORM = Full.

\$CDETAIL	NTWKNUM
\$CHKPNT	OP
\$MODESEL	PLCADDR
\$PTRESID	PNTFORM
ASSOCDSP	PNTSTATE
BOXCLR	PNTTYPE
DISPTYPE	PTDESC
DOTYPE	PTEXECST
EUDESC	REDTAG(F)
KEYWORD	S0BOXCLR
MODATTR(F)	S1BOXCLR
MODE(F)	SLOTNUM
MODEAPPL(F)	SO
MODEPERM(F)	STATE0-1
NMODATTR(F)	STATETXT
NMODE(F)	UNITNUM
NODENUM	
NODETYP	

2.10 LOGIC

The parameters of the Logic Data Point (otherwise referred to as the Logic Slot) are listed below in alphabetical order. (F) indicates that the parameter is applicable when the PNTFORM = Full.

\$CDETAIL	NOLINPTS
\$CHKPNT	NOLOGBLK
\$PRMDESC(F)	NOLOPTS
\$PTRESID	NTWKNUM
ASSOCDSP	PNTFORM
DISPTYPE	PNTSTATE
GENDESC(F)	PNTTYPE
L	PRIMMOD
LIBADOPT	PRMDESC(F)
LISRC	PTEXECST
LODSTN	PVFLTPT
LOENBL	SLOTNUM
LOGMIX	UNITNUM
NODENUM	
NODESC(F)	
NODETYP	

2.11 UCN

The parameters of the UCN Data Point are listed below in alphabetical order.

BADPVTXT	NMSGTXT
LOADSCOP	NODENUM
MOVPVTXT	NODETYP
MSGTXT	NTWKNUM
NODEASSN	

**\$ALDLOPT**

Type: **E: \$ALDLOPT** This R620 parameter, in conjunction with the DLYTIME parameter, is
Lock: **Eng** intended to reduce the number of unnecessary alarms reported by analog input
Default: **None** points during period when the process is unstable
PtRes: **NIM** (startup/shutdown/upsets/spikes).

The \$ALDLOPT parameter can be used to implement an on-delay or an off-delay. The off-delay option delays turning alarms off for the time period specified by the DLYTIME parameter to minimize the effects of an input value oscillating in and out of alarm. The on-delay options delays turning alarms on for the time period specified by the DLYTIME parameter to suppress the reporting of transient alarms, or to suppress alarms which are not important to the operator unless they stay in alarm for a minimum amount of time.

The time of reporting of the alarm/return-to-normal events, which underwent alarm delay, will be the time when the delay timer has expired, not the actual time of occurrence of the event.

Refer to the definition of the DLYTIME parameter for the time-delay value.

WARNING: Out-of-the-ordinary situations and configurations can cause peculiar results. Therefore, the alarm delay option with its alarm priority filter should be used in such a way that critical safety related alarms will not be suppressed or delayed by the alarm delay option. The effect of alarm delay in different scenarios should be understood before configuring any point with this option.

Range: **None:** Neither on delay nor off delay is selected.

OffDlyLo: Off delay selected and alarms with less than or equal to Low priority are subjected to alarm delay functionality.

OffDlyHi: Off delay selected and alarms with less than or equal to High priority are subjected to alarm delay functionality.

OffDlyEm: Off delay selected and alarms with less than or equal to Emergency priority are subjected to alarm delay functionality.

OnDlyLo: On delay selected and alarms with less than or equal to Low priority are subjected to alarm delay functionality.

OnDlyHi: On delay selected and alarms with less than or equal to High priority are subjected to alarm delay functionality.

OnDlyEm: On delay selected and alarms with less than or equal to Emergency priority are subjected to alarm delay functionality.

\$AUXUNIT

\$AUXUNIT

Type: **E:UNIT** **Auxiliary Unit**—The Auxiliary unit of an alarmable process point. If an Auxiliary Unit ID is specified, alarms from this point go to the Auxiliary Unit instead of the Primary Unit. If the \$AUXUNIT parameter is set to null (- -), alarms go to the Primary Unit. Available in Release 520 and later software..

Lock: **Oper**

Default: **Null**

PtRes: **NIM**

Range: **A-Z, 0-9, and Null** – (Any valid unit ID as configured in the NCF).

<i>Helpful Hint:</i> A Network configuration option sets the key level required to change \$AUXUNIT.
--

\$CDETAIL

Type: **String** **Custom Detail Display Name**—This R620 parameter contains the 8-character name of the custom schematic that defines the user-supplied Custom Detail Display. When the “Detail” of the point is requested, the schematic identified by the \$CDETAIL parameter is invoked instead of the standard Detail Display. If the \$CDETAIL parameter is blank, the standard Detail Display call is made. If the \$CDETAIL parameter is not blank, and the schematic specified in that parameter cannot be invoked, then the standard Detail Display will be invoked with an error message that informs the user that the invocation of the Custom Detail Display failed.

Lock: **Eng**

Default: **Blank**

PtRes: **NIM**

Range: **N/A**

\$CHKPNT (All Point Types)

Type: **Blind Record** **Checkpoint Record**—This is a special database save function.

Lock: **IntUsr**

Default: **None**

PtRes: **LMM**

Range: **None**

\$COMCKPT (Node Specif.)

Type: **Blind Record** **LM Node Specific Data Point Checkpoint Record**

Lock: **IntUsr**

Default: **None**

PtRes: **LMM**

Range: **None**

\$LADDER(1)–(250) (Node Specif.)

Type: **Blind Record** **An LC program**—This program has an array of up to 250 ladder logic instructions.

Lock: **IntUsr**

Default: **None**

PtRes: **LMM**

Range: **1 to 250**

\$MODESEL (AnalgOut, DigOut, DigComp)

Type: **Logical, Array**
(1...5)

Lock: **Eng**

Default: **On {mode is allowable}**

PtRes: **NIM**

Mode Selection List — This R630 parameter allows the engineer to remove the MODE/MODATTR selections from the operator's standard-display MODE selection lists. This prevents the data point from being placed in the wrong mode and from having the wrong mode attribute assigned.

This function is implemented by the user configuring a mode selection list to determine the applicable modes and mode attribute. During normal operation, the mode selection list is displayed on the standard detail, faceplate, group, and change zone displays when you select the mode parameter. The operation of the MAN/AUTO/NORM keyboard keys is also affected by the mode selection list choices.

Range:

- \$MODESEL(1) - On Allows the user to select Man mode from a display.
- Off Does not allow the user to select Man mode from a display.
- \$MODESEL(2) - Not used.
- \$MODESEL(3) - On Allows the user to select Cas mode from a display.
- Off Does not allow the user to select Cas mode from a display.
- \$MODESEL(4) - Not used.
- \$MODESEL(5) - On Allows the user to select Prog mode attribute from a display.
- Off Does not allow the user to select Prog mode attribute from a display.

\$OPTOL (AnagOut)

Type: **AnalgOut**

Lock: **Eng**

Default: **0.0**

PtRes: **NIM**

Output Tolerance — Tolerance limit for manually entered OP. The difference between a new OP and a current OP is compared against \$OPTOL. If the tolerance is violated in either a positive or negative direction from the current value of OP, operator confirmation is required before the value is stored. A value of 0.0 disables this check. A NaN or a negative value is not allowed.

Range: **0.0 to 106.9%**

\$PAISLOT(xx) (AnalgIn)

Type: **Integer**

Lock: **View**

Default: **0**

PtRes: **NIM**

Range: **>=0**

Parameter Analog Input Slot – Indicates the number of Analog Input points configured on the selected node.

\$PAOSLOT(xx) (AnalgOut)

Type: **Integer**

Lock: **View**

Default: **0**

PtRes: **NIM**

Range: **>=0**

Parameter Analog Output Slot – Indicates the number of Analog Output points configured on the selected node.

\$PDCSLOT(xx)

\$PDCSLOT(xx) (DigComp)

Type: **Integer** **Parameter Digital Composite Slot** – Indicates the number of Digital Composite points configured on the selected node.
Lock: **View**
Default: **0**
PtRes: **NIM**
Range: **>=0**

\$PDISLOT(xx) (DI)

Type: **Integer** **Parameter Digital Input Slot** – Indicates the number of Digital Input points configured on the selected node.
Lock: **View**
Default: **0**
PtRes: **NIM**
Range: **>=0**

\$PDOSLOT(xx) (DO)

Type: **Integer** **Parameter Digital Output Slot** – Indicates the number of Digital Output points configured on the selected node.
Lock: **View**
Default: **0**
PtRes: **NIM**
Range: **>=0**

\$PLOGSLT(xx) (Logic)

Type: **Integer** **Parameter Logic Slot** – Indicates the number of Logic points configured on the selected node.
Lock: **View**
Default: **0**
PtRes: **NIM**
Range: **>=0**

\$PRMDESC (Logic)

Type: **String_8** **Parameter Descriptor Assignment** — Defines up to 12 logic slot parameters to which custom generic descriptors entered through parameters GENDESC (1-12) are to be assigned.
Lock: **View**
Default: **None**
PtRes: **NIM**
Range: **L1..L12 (Logic slot inputs)**
 FL1..FL12 (Logic slot flags)
 NN1..NN4 (Logic slot numerics)
 SO1..SO16 (Logic slot outputs)

\$PTIME(xx) (Timer)

Type: **Integer** **Parameter Timer** – Indicates the number of Timer points configured on the selected node.
Lock: **View**
Default: **0**
PtRes: **NIM**
Range: **>=0**

\$PTRESID (All Point Types)

Type: **Blind Record** **A Point Descriptor— Point Residency Identifier.**
Lock: **View**
Default: **Blank**
PtRes: **NIM**
Range: **None**

\$PUANLIN(xx) (AnalgIn)

Type: **Real** **PUs consumed by Analog Input point** – Indicates the number of PUs
Lock: **View** consumed by the configured Analog Input point for the selected node.
Default: **0.0**
PtRes: **NIM**
Range: **>=0**

\$PUANLOP(xx) (AnalgOut)

Type: **Real** **PUs consumed by Analog Output point** – Indicates the number of PUs
Lock: **View** consumed by the configured Analog Output point for the selected node.
Default: **0.0**
PtRes: **NIM**
Range: **>=0**

\$PUDIGCP(xx) (DigComp)

Type: **Real** **PUs consumed by Digital Composite point**– Indicates the number of PUs
Lock: **View** consumed by the configured Digital Composite point for the selected node.
Default: **0.0**
PtRes: **NIM**
Range: **>=0**

\$PUDIGIN(xx) (DI)

Type: **Real** **PUs consumed by Digital Input point** – Indicates the number of PUs
Lock: **View** consumed by the configured Digital Input point for the selected node.
Default: **0.0**
PtRes: **NIM**
Range: **>=0**

\$PUDIGOP(xx) (DO)

Type: **Real** **PUs consumed by Digital Output point** – Indicates the number of PUs
Lock: **View** consumed by the configured Digital Output point for the selected node.
Default: **0.0**
PtRes: **NIM**
Range: **>=0**

\$PUFREE

\$PUFREE(xx) (Node Specif.)

Type: **Real** **Free PUs** – Indicates the number of PUs available in the selected node.
Lock: **View**
Default: **0.0**
PtRes: **NIM**
Range: **>=0**

\$PULOGIC(xx) (Logic)

Type: **Integer** **PUs consumed by Logic point** – Indicates the number of PUs consumed by
Lock: **View** the configured Logic point for the selected node.
Default: **0**
PtRes: **NIM**
Range: **>=0**

\$PUTIME(xx) (Timer)

Type: **Real** **PUs consumed by Timer point** – Indicates the number of PUs consumed by
Lock: **View** the configured Timer point for the selected node.
Default: **0.0**
PtRes: **NIM**
Range: **>=0**

\$PVSTREC (AnalIn)

Type: **Blind Record** **PV Value and Status Record**—A special request that retrieves the PV Value
Lock: **View** and the Status of the PV Input Value (PVSTS).
Default: **None**
PtRes: **NIM**
Range: **N/A**

\$RECOVER (Node Specif.)

Type: **Boolean** A write-only parameter which initiates recovery of all LMM alarms when
Lock: **IntUsr** written to True/On.
Default: **None**
PtRes: **LMM**
Range: **True**

\$REALARM

\$REALARM (AnalogIn, DigComp, DigIn, Flag)

Type: **Integer** **Re-alarms Interval** — With R660, acknowledged and unacknowledged process alarms (event-type 21) are rebroadcast if they have not returned to normal (RTN) within the re-alarms interval specified by the \$REALARM parameter. The re-alarm feature is applicable for all types of process alarms that can have an RTN event. The user-configured re-alarm interval is the same value for all process alarms associated with the same point.

Lock: **Eng**

Default: **0 (no re-alarm)**

PtRes: **NIM**

Re-alarms is consistent across all stations in the console in that area.

Rebroadcast alarms are also sent to the History Module with a new timestamp, and reported to the RTJ and Process Alarm Journal where they will appear as an alarm.

On the alarm displays, the rebroadcast alarm behaves like a new alarm but has a yellow time stamp.

Rebroadcast alarms appear on the GUS Alarm Summary, and on the following GUS/US native-window displays:

- Area Alarm Summary
- Unit Alarm Summary
- Alarm Annunciator
- Point Detail

The re-alarm feature is implemented using the \$REALARM parameter that has a range of 0 through 9999 minutes. The value specified for this parameter determines the interval, in minutes that will elapse before the specified point is re-alarmed.

Range: **0 through 9999 minutes** – (value range is from one minute to approximately 167 hours). A value of 0 minute disables the rebroadcast alarm function.

-A-

ALENBST (AnalIn, DigComp, DigIn, Flag)

Type: E:ALENBST **Alarm Enable Status**—Defines the alarm reporting function that is to be used when an alarm condition is detected in this data point.

Lock: See Note

Default: Enable

PtRes: NIM

Range:	<u>Displayed</u>	<u>Logged</u>	<u>Reported to EIP</u>
Enable	Yes	Yes	Yes
Disable*	Yes	Yes	Yes
Inhibit	No	No	No

*Disabled alarms are not annunciated at Universal Stations, but the alarm indicators appear on the Group and Detail displays.

Helpful Hint: ALENBST should not be set to Disable or Inhibit for points critical to safe operations. For Node Specific Flag points, this parameter applies to only slots 1 through 128.

NOTE

The access lock for ALENBST can be configured in Network Configuration under System-Wide Values/Console Data to Engr, Supr, or Oper.

ALMOPT (DigComp)

Type: E:\$ALMOPT **Alarm Option**—Defines the types of alarms that can be detected by this data point. Refer to the *LM Control Functions* manual for a detailed description.

Lock: Eng/PB

Default: None

PtRes: LMM

Range: None (No alarms are to be detected)

Offnorml (Off Normal; alarm if PV and PVNORMAL are not the same state.)

Cmddis (Command Disagree; alarm if field device did not respond to commanded state within configured feedback time; also detects uncommanded changes in the field device. Digital Composite only.)

Helpful Hint:

1. Offnorml alarm can be configured only if the point has at least one input.
2. Cmddis (and uncommanded change) alarm can be configured only if the point has at least one input and one output.

ALMOPT

ALMOPT (DigIn)

Type: E:\$ALMOPT **Alarm Option**—Defines the alarming option for a digital input point whose
Lock: Eng/PB DITYPE is Status.
Default: None
PtRes: LM
Range: None (No alarms are to be detected.)
Offnorml (Off Normal; alarm if current PV state is not the PVNORMAL state. PVNORMAL is defined by the STATETXT(0) or STATETXT(1) descriptor, as configured by the user.)
ChngofSt (An alarm is generated when the digital input changes state in either direction.)

<i>Helpful Hint:</i> ALMOPT configuration requires DITYPE = Status.

ALPRIOR (DigIn, DigComp, Flag)

Type: E:ALPRIOR **Alarm Priority**—Defines the alarm priority for digital and flag points.
Lock: Eng/PB
Default: Low
PtRes: NIM
Range: Emergency (Reported to all alarm summary displays)
High (Reported to Area Alarm Summary Display and Unit Alarm Summary Display)
Low (Reported to Unit Alarm Summary Display)
Journal (Logged, indicators appear on Group and Detail displays)
NoAction (Alarm is not reported to the system)

<i>Helpful Hint:</i> For Node Specific Flag points, this parameter applies to only slots 1 through 128.

ASSOCDSP

Type: String_8 **Associated Display**—Specifies a user-configured schematic that is associated
Lock: Eng with this point. Available on Release 510 and later software.
Default: Blank
PtRes: NIM
Range: N/A

<i>Helpful Hint:</i> The specified associated display can be called from a Point Detail Display, or from any summary display, or the Group display when the point is selected.
--

-B-**BADCTLFL (DigComp)**

Type: **Boolean** **Bad Control Flag**—If the calculated Logic Controller output state in OPFINAL does not match the commanded state in OP, BADCTLFL = On, generating a
Lock: **View**
Default: **Off** Bad Control Alarm; and on the Group Detail Display for this point, the OP state is shown as Undefined, and the Bad Control alarm indicator (BC) appears.
PtRes: **LMM** Where more than one LC output for this point is in use, the LMM compares each output with OP, and reports the first match in OPFINAL. The Bad Control alarm is generated if no match is found on any output.

Range: The permissible character set for the state descriptors are as follows:
 Alphabetic, A-Z (upper case only)
 Numerals 0-9
 Underscore (_)

BADCTLPR

Type: **E:ALPRIOR** **Bad Control Alarm Priority**—Defines the priority of the bad control alarm.
Lock: **Eng**
Default: **NoAction** With R670, the default value is changed to NoAction.
PtRes: **NIM**
Range: **JnlPrint** (Alarm is historized and reported to the printer, but not annunciated.) **Printer**
 (Alarm is reported to the printer, but not historized and not annunciated.) **Emergency**
 (Alarm is historized, annunciated, and reported to all alarm summary displays.)
High (Alarm is historized, reported to Area Alarm Summary Display and Unit Alarm Summary Display.)
Low (Alarm is historized, reported to the Unit Alarm Summary Display, and annunciated.)
Journal (Alarm is historized, but not reported to Universal Stations and not annunciated.)
NoAction (Alarm is not reported to the system and not annunciated.)

BADPVFL (AnalogIn, DigComp, DigIn)

Type: **Logical** **Bad PV Flag**—Indicates that a bad PV value has been detected at this data
Lock: **View** point. For an analog input, a bad PV is defined as a PV whose value is NaN
Default: **Off** (AnalogIn, (Not a Number).
 DigComp)
 On (DigIn)
PtRes: **LMM**
Range: **Off** (PV is not bad)
 On (PV is bad)

BADPVPR

Type: **E:ALPRIOR** **Bad PV Alarm Priority**—Defines the priority of the bad PV alarm.
Lock: **Eng/PB**
Default: **NoAction** With R670, the default value is changed to NoAction.
PtRes: **NIM**
Range: **JnlPrint** (Alarm is historized and reported to the printer, but not annunciated.) **Printer** (Alarm is reported to the printer, but not historized and not annunciated.) **Emergency** (Alarm is historized, annunciated, and reported to all alarm summary displays.) **High** (Alarm is historized, reported to Area Alarm Summary Display and Unit Alarm Summary Display.) **Low** (Alarm is historized, reported to the Unit Alarm Summary Display, and annunciated.) **Journal** (Alarm is historized, but not reported to Universal Stations and not annunciated.) **NoAction** (Alarm is not reported to the system and not annunciated.)

BADPVTEXT (Node Specif., UCN Node)

Type: **String_8** **Bad PV State Descriptor**—Defines the state descriptor that is displayed when the digital composite point state is indeterminate or bad. The bad state can result when the PV input signals from the process are in an inconsistent state (e.g., for a valve, the limit switches indicating open and closed are on at the same time). This state descriptor is configured once and applies to all digital composite points in this LMM.
Lock: **PtBld**
Default: **BAD**
PtRes: **NIM**

Range: The permissible character set for the up to eight character descriptor is as follows:
 Alphabets A-Z (upper case only)
 Numerics 0-9
 Underscore (_)

Helpful Hint: If NODENUM is an even number, this parameter is hidden on the UCN Node Configuration Display.

BOXCLR

BOXCLR (DigComp)

Type: E:BOXCOLOR
Lock: Eng/PB

Box Color for Digital Displays—Defines the color of the upper, middle, and lower boxes that are used to display the current state of the point on the Group and Detail Displays. The lower box and its default (Yellow) do not apply if NOSTATES = 2 for this point.

Default: Red [Upper box default (State 1)]
Green [Middle box default (State 0)]
Yellow [Lower box default (State 2)]
PtRes: NIM
Range: Red
Green
White
Black
Cyan
Yellow
Blue
Magenta

<i>Helpful Hint:</i> BOXCLR has an access lock of View if PNTFORM = Component.
--

BOXCLR (DigIn, Flag)

Type: E:BOXCOLOR
Lock: Eng/PB

Box Color for Digital Displays—Defines the color of the upper and lower boxes that are used to display the current state of the point on the Group and Detail Displays. Boxes are applicable when DITYPE is Latched or Status. Access lock will be View if PNTFORM = Component.

Default: Green [Upper box default color (State 1)]
Yellow [Lower box default color (State 0)]
PtRes: NIM
Range: Red
Green
White
Black
Cyan
Yellow
Blue
Magenta

<i>Helpful Hint:</i> BOXCLR has an access lock of View if PNTFORM = Component.
--

BYPASS

BYPASS (DigComp)

Type: **Logical** **Override Input Bypass Enable**—Allows a store to the OP and the SO
Lock: **Oper** parameters that ignores interlocking if “ON”.
Default: **Off**
PtRes: **LMM**
Range: **Off** (Bypass of inputs is not allowed)
 On (Bypass of inputs is allowed)

-C-**CASREQ (AnalogOut)**

Type: E:CASREQ **Remote Cascade Request Flag**—Defines whether the remote cascade mode has been requested for the data point. The remote cascade mode exists when MODE is changed to Cas and RCASOPT is Ddc or DdcRsp. When a request to change MODE to Cas is received from a US or a program, MODE does not immediately change to Cas. Instead, CASREQ is set to Request and a -C appears to the right of the mode indicator on the Group and Detail displays. When continuous control in an AM determines that CASREQ contains Request, it requests the mode to go to Cas, and the LMM changes CASREQ to NotReq.

Lock: Prog

Default: NotReq

PtRes: LMM

Range: 0-**NotReq** (Remote cascade mode request not made.)

1-**Request** (Remote cascade mode request made; operator or program has requested the cascade mode.)

Helpful Hint: CASREQ does not apply for an AnalogOut point if RCASOPT = None. Ddc is the only remote cascade option for an analog output point.

CCSRC (AnalogIn, DigComp, DigIn, Flag)

Type: Integer **Contact Cut Out**—Source address for a boolean value that determines the state of the CONTCUT parameter. When it is unconfigured, its value is -1, and CONTCUT may be written from another UCN node.

Lock: PtBld

Default: -1

PtRes: LMM

Range: -1 to 8191

Helpful Hint: For LM Node Specific Flag points, CUTOOUT applies to only slots 1–128.

CHPINOPR (Node Specif.)

Type: E:CHPINDAC **Automatic Checkpoint Inhibit Operation**—Defines whether automatic database saves are to be performed for the devices connected to this NIM.

Lock: Oper

Default: Enable

PtRes: NIM

Range: **Enable** (Automatic database saves are enabled)

Disable (Automatic database saves are disabled)

CMDDISFL (DigComp)

Type: **Logical** **Command Disagree Alarm Flag**—Indicates whether a field device did not go to the commanded state within the allowed feedback time.
Lock: **View**
Default: **Off**
PtRes: **LMM**
Range: **Off** (No command disagree alarm)
On (Command disagree alarm has been detected by this point)

Helpful Hint: A slow-responding field device can cause a premature alarm. If so, adjust the time in parameter FBTIME.

CMDDISPR

Type: **E:ALPRIOR** **Command Disagree Alarm Priority** - Defines the alarm priority of command disagree, command fail, and uncommanded change alarms.
Lock: **Eng**
Default: **NoAction** With R670, the default value is changed to NoAction.
PtRes: **NIM**
Range: **JnlPrint** (Alarm is historized and reported to the printer, but not annunciated.) **Printer** (Alarm is reported to the printer, but not historized and not annunciated.) **Emergency** (Alarm is historized, annunciated, and reported to all alarm summary displays.) **High** (Alarm is historized, reported to Area Alarm Summary Display and Unit Alarm Summary Display.) **Low** (Alarm is historized, reported to the Unit Alarm Summary Display, and annunciated.) **Journal** (Alarm is historized, but not reported to Universal Stations and not annunciated.) **NoAction** (Alarm is not reported to the system and not annunciated.)

COMMAND (Timer)

Type: **E:COMMAND** **Timer Commands**—Allow the operator to control the operation of the timer data point.
Lock: **Oper**
Default: **None**
PtRes: **LMM**
Range: **0-None** (No effect on the timer)
1-Start (Starts the timer)
2-Stop (Stops the timer)
3-Reset (Resets the timer to zero)
4-RestStrt (Resets the timer, then starts the timer)

CONTCUT

CONTCUT (AnalogIn, DigComp, DigIn, Flag)

Type: **Logical** **Contact Cut Out**—Defines whether alarms detected at this data point are to be cut out to prevent this data point's alarms from being reported to the operator.

Lock: **Prog** The alarms continue to be reported to the journals and the AM or CM through the EIPPCODE parameter.

Default: **False for all except Flag, where default is Off.** CONTCUT can be used to cutout alarms on a point when the alarms are generated because of specific conditions at other points which themselves have alarms. As an example, the user could configure a logic point so that the logic point would monitor the nuisance alarm conditions and then store the contact cutout state of this point using an output connection. It can also be stored by the sequence program in the LM or the AM, which could monitor the process conditions to determine when the alarms have to be suppressed.

PtRes: **LMM**

If CCSRC is -1, CONTCUT is writable from the UCN. In addition, if CCRSC is a valid LC address, CONTCUT would be locked at View only, and would equal the value of the LC boolean value of CCSRC.

Range: **True** (Analog Input, Digital Composite, and Digital Input. **On** for LM Node Specific Flag Points.)
False (Analog Input, Digital Composite, and Digital Input. **Off** for LM Node Specific Flag Points.)

Helpful Hint: Also causes this point's alarms to be journaled and reported to LCN modules but not to be reported to the Alarm Summary Display at the Universal Station. For LM Node Specific Flag points, CUTOOUT applies to only slots 1–128.

CRPPXORN (Node Specif.)

Type: **Integer** **Current Hour Point Processing Overrun Counter**—A counter that accumulates and shows the number of LMM point processing overruns that have occurred during the current hour.

Lock: **View**

Default: **0**

PtRes: **LMM**

Range: **≥ 0**

CRUNCORN (Node Specif.)

Type: **Integer** **Current Hour Point Processing Fetch/Store Counter**—A counter that accumulates and shows the number of LMM point processing fetches or stores that have occurred during the current hour.

Lock: **View**

Default: **0**

PtRes: **LMM**

Range: **≥ 0**

-D-

D1, D2 (DigComp)

Type: Logical **Digital Input 1 Status and Digital Input 2 Status**—Separately indicate whether input 1 and input 2 are on or off.
Lock: View
Default: Off
PtRes: LMM
Range: Off (Input inactive)
 On (Input is active)

D1_0 (DigComp)

Type: E:\$PVSTATS **Digital Input 1 Equal to a PV State of 0**—Defines the PV state that corresponds to input D1 = Off.
Lock: View
Default: PVState0
PtRes: LMM
Range: PVState0 (STATETXT0 describes D1 = 0)
 PVState1 (STATETXT1 describes D1 = 0)

Helpful Hint: Applies only if NODINPTS = 1. D1_0 is always the opposite state of D1_1.

D1_1 (DigComp)

Type: E:\$PVSTATS **Digital Input 1 Equals a PV State of 1**—D1_1 defines the PV state that corresponds to D1 (Input 1 status) = On.
Lock: Eng/PB
Default: PVState0
PtRes: LMM
Range: 0-PVState0 (STATETXT0's content describes D1 = 1)
 1-PVState1 (STATETXT1's content describes D1 = 1)

Helpful Hint: D1_1, Digital Input 1 Equal to a PV State of 1, applies only if NODINPTS = 1. D1_1 is always the opposite state of D1_0 and vice versa.

D2D1_00 (DigComp)

Type: E:\$PVSTATS **D2_D1 Zero_Zero PV State**—Defines the PV state descriptor that is to be used and displayed when inputs D2 and D1 are both Off (00).
Lock: Eng/PB
Default: MovPV
PtRes: LMM
Range: 0-PVState0 (STATETXT0 descriptor)
 1-PVState1 (STATETXT1 descriptor)
 2-BadPV (BADPVTXT descriptor)
 3-MovPV (MOVPVTXT descriptor)
 4-PVState2 (STATETXT2 descriptor)

Helpful Hint: D2D1_00 configuration requires NODINPTS=2. Option PVState2 cannot be selected unless NOSTATES = 3. STATETXT(0–2) is configured for each DigComp point; BADPVTXT and MOVPVTXT are configured during Node Specific Data Point configuration for all DigComp points in this LMM.

D2D1_01 (DigComp)

Type: E:\$PVSTATS **D2D1 Zero_One PV State**—Defines the PV state descriptor that is to be used and displayed when input D2 is Off and input D1 is On (01).
Lock: Eng/PB
Default: PVState0
PtRes: LMM
Range: PVState0 (STATETXT0 descriptor)
PVState1 (STATETXT1 descriptor)
PVState2 (STATETXT2 descriptor)
BadPV (BADPVTXT descriptor)
MovPV (MOVPVTXT descriptor)

Helpful Hint: D2D1_01 configuration requires NODINPTS = 2. Option PVState2 cannot be specified unless NOSTATES = 3. STATETXT(0–2) is configured for each DigComp tag name; BADPVTXT and MOVPVTXT are configured during Node Specific Data Point configuration for all DigComp points in the LMM.

D2D1_10 (DigComp)

Type: E:\$PVSTATS **D2D1 One_Zero PV State**—Defines the PV state descriptor that is to be used and displayed when input D2 is On and input D1 is Off (10).
Lock: Eng/PB
Default: PVState1
PtRes: LMM
Range: 0-PVState0 (STATETXT0 descriptor)
1-PVState1 (STATETXT1 descriptor)
2-BadPV (BADPVTXT descriptor)
3-MovPV (MOVPVTXT descriptor)
4-PVState2 (STATETXT2 descriptor)

Helpful Hint: D2D1_10 configuration requires NODINPTS = 2. Option PVState2 cannot be specified unless NOSTATES = 3. STATETXT(0–2) is configured for each DigComp point; BADPVTXT and MOVPVTXT are configured during Node Specific Data Point configuration for all DigComp points in the LMM.

D2D(1)–(11) (DigComp)

Type: E:\$PVSTATS **D2D1 One_One PV State**—Defines the PV state descriptor that is to be used and displayed when inputs D2 and D1 are both On (11).
Lock: Eng/PB
Default: BadPV
PtRes: LMM
Range: 0-PVState0 (STATETXT0 descriptor)
1-PVState1 (STATETXT1 descriptor)
2-BadPV (BADPVTXT descriptor)
3-MovPV (MOVPVTXT descriptor)
4-PVState2 (STATETXT2 descriptor)

Helpful Hint: D2D1_11 configuration requires NODINPTS = 2. Option PVState2 cannot be specified unless NOSTATES = 3. STATETXT(0–2) is configured for each DigComp point; BADPVTXT and MOVPVTXT are configured during Node Specific Data Point configuration for all DigComp points in the LMM.

DISPTYPE

DISPTYPE (Analg I/O, Dig I/O, DigComp, Flag, Logic, Numeric, Timer)

Type: **E:\$DISPTYPE** **Display Type**—Defines, for the Operator Personality, the type of Detail and

Lock: **View** Group display present to represent this point.

Default: **Null**

PtRes: **NIM**

Range: **See Chart Below**

Point Type	Range
Analog Input	AnalgIn Null
Analog Output	AnalgOut Null
Digital Input	DigIn Null
Digital Output	DigOut Null
Digital Composite	DigComp Null
Logic	Logic Null
Numeric	Numeric Null
Flag	Flag Null
Timer	Timer Null

DISRC(1)–DISRC(2) (DigComp)

Type: **Universal** **Digital Composite Input-Connection Source**—Specifies the sources whose values are to be fetched and delivered to Digital Composite data point inputs D1 and D2. The source can be specified using the "Tagname.Parameter" format or the hardware reference address format. Refer to the *LM Control Functions* manual for a detailed description.

Lock: **PtBld**

Default: **Null**

PtRes: **LMM**

Range: Use Tagname.Parameter format for tagged points where Tagname can be up to 8, or up to 16 characters (depending on the system option) and the permissible character set is as follows:
 Alphabetics A-Z (upper case only).
 Numerics 0-9 (an all numeric tag name is not allowed).
 Underscore (_) cannot be used as the first character or the last character, and consecutive underscores are not allowed.
 Embedded space characters are not allowed.
 An * is used to default to this point's tag name.
 Parameter name can be up to eight characters, and must be a legitimate parameter name.

Some possible input-connection sources are:

- a. "DigIn slot Tagname.PVFL"
- b. "Node Specific Flag slot Tagname.PVFL(nnnn)" where nnnn = 1–1024
- c. "!BoxXX.FL(nnnn)" for a node specific flag that resides in the same LMM; XX = LMM node specific number, and nnnn = 1–1024.
- d. "\$NMhhBxx.FL(nnnn)" for a node specific flag that resides in the same LMM; hh is the NIM UCN address, xx is the LMM node specific number, and nnnn = 1–1024.

Use the hardware reference address !ADRXXXX.FL for direct connection to the LC.

DITYPE (DigIn)

Type: **E:\$DITYPE** **Digital Input Type**—Defines the type of digital input point.

Lock: **PtBld**

Default: **Status**

PtRes: **LMM**

Range: **0-Status** (Point is to be used for alarming and event reporting)
 (The values of 1-Latched and 2-Accumulated apply to PMs and APMs only; they do not apply to Logic Managers.)

DLYTIME

DLYTIME (AnalogIn)

Type: **Integer**

Lock: **Eng**

Default: **0 seconds**

PtRes: **NIM**

Delay Time—(R620) (Analog Input) This R620 parameter enables suppression of redundant alarms caused by chattering analog values.

This parameter specifies the amount of time during which the alarm reporting (in the case of on delay) and return-to-normal reporting (in the case of off delay) is prevented. Alarms that return to normal during the time delay will not be reported. The timer will be restarted on either every transition into alarm (for on delay) or for every transition out of alarm (for off delay).

Range: **0 to 999 seconds**

Warning: This parameter can be accessed through DATCHNG, Point Detail Display, DEB or AM/CL. PM/CL cannot access DLYTIME parameter since it is resident in NIM.

NOTE

- a NaN or negative value is not allowed.
- EIP functionality is not affected by the Delay Option.

DLYTIME (DigIn)

Type: **Integer**

Lock: **Supr**

Default: **0 seconds**

PtRes: **LMM**

Range: **0 to 60 seconds**

Delay Time—Defines the time (in seconds) that a detected alarm is guaranteed to remain in alarm even if the condition clears.

Helpful Hint: DLYTIME (DigIn) configuration requires ALMOPT = Offnorm.

DODSTN(1)–(3) (DigComp)

Type: **Universal** **Digital Composite Output-Connection Destination**—Specifies up to three output connection destinations that are to receive the OP output from this digital composite point. The destination can be specified using the "Tagname.Parameter" format or the hardware reference address format. Refer to the *LM Control Functions* manual for a detailed description.

Lock: **PtBld**

Default: **Null**

PtRes: **LMM**

Range: Use Tagname.Parameter for tagged points where Tagname can be up to 8, or up to 16 characters (depending on the system option) and the permissible character set is as follows:
 Alphabetic A-Z (upper case only).
 Numerics 0-9 (an all numeric tag name is not allowed).
 Underscore (_) cannot be used as the first character or the last character, and consecutive underscores are not allowed.
 Embedded space characters are not allowed.
 An * is used to default to this point's tag name.
 Parameter name can be up to eight characters, and must be a legitimate parameter name.

Some possible output-connection destinations are:

- a. "DigOut slot Tagname.SO"
- b. "NodespecificXX.FL(nnnn)" for a node specific flag that resides in the same LMM; where XX = LMM node specific number, and nnnn = 1–1024.
- c. "\$NMhhBxx.FL(nnnn)" for a node specific flag that resides in the same LMM; hh is the LMM UCN address, xx is the LMM node specific number, and nnnn = 1–1024.

Use the hardware reference address !ADRXXXX.FL for direct connection to the LC.

DOTYPE (DigOut)

Type: **E:\$DOTYPE** **Digital Output Type**—Determines the type of digital output point.

Lock: **PtBld**

Default: **Status**

PtRes: **LMM**

Range: **0-Status** (Status output type)

-E-**EIPPCODE (DigComp, DigIn, Flag)**

Type: **Ent_Id** **Event-Initiated Processing Point Identifier**—Defines the tag name of the point in an AM or CM that is to be notified when an event is detected by this point.

Lock: **PtBld**

Default: **Null**

PtRes: **NIM**

Range: Tag name of the data point can be up to 8, or up to 16 characters (depending on the system option) and the permissible character set is as follows:
 Alphabets A-Z (upper case only)
 Numerics 0-9 (an all numeric tag name is not allowed)
 Underscore (_) cannot be used as the first character or the last character, and consecutive underscores are not allowed.
 Embedded space characters are not allowed.

Helpful Hint: EIPPCODE configuration requires PNTTYPE = DigIn or DigComp and EVTOPT = Eip or EipSoe. For LMM Node Specific Flag points, this parameter applies only to slots 1 through 128.

EUDESC (Analog I/O, Dig I/O, DigComp, Flag, Numeric, Timer)

Type: **String_8** **Engineering Units Descriptor**—An eight character descriptor that defines the name of the engineering units (EU) that are displayed on the Group and Detail

Lock: **PtBld**

Default: **Blank** Displays for this point as shown in Figure N-1 (see NAME). In this figure,

PtRes: **NIM** LBS/SEC is the engineering unit descriptor.

Range: Permissible character set consists of all characters on the Engineer's Keyboard. This set consists of alphabets A-Z, numerics 0-9, and the following special characters: space ! " % & ' () * + - / : ; > < = ? _ , . \$

EVTOPT (DigComp, DigIn)

Type: **E:\$EVTOPT** **Event Reporting Option**—If EVTOPT = Eip and a PV alarm is generated, the AM or CM data point named EIPPCODE is notified and a "process special" on that data point takes place.

Lock: **PtBld**

Default: **None**

PtRes: **LMM**

Range: **None** (Event-Initiated Processing is not allowed)
Eip (Process special is triggered in AM/CM)

Helpful Hint: EVTOPT configuration requires NODINPTS > 0.

-F-**FBTIME (DigComp)**

Type: **Integer** **Feedback Time**—Sets the amount of time (in seconds) that the point should wait before generating a "command disagree" alarm after the operator has issued a start/stop-type command to a field device.

Lock: **Eng/PB**

Default: **1000**
seconds

PtRes: **LMM**

Range: **1 to 1000 seconds**

Helpful Hint: FBTIME can be increased to compensate for a slow-responding field device that does not respond to the operator's command in time to prevent a command-disagree alarm.

FINSTTXT(1)–(3) (DigComp)

Type: **String_8** **Final Output State Descriptor**—An array of four descriptor strings, one for each of the four possible output states. The descriptor configured for the output state read back from the Logic Controller becomes the state shown for OP for this point on the Group and Detail displays. Because the commanded state can differ from the final output state from the Logic Controller, the text in FINSTTXT can differ from the text in OPSTTEXT.

Lock: **View**

Default: **N/A**

PtRes: **NIM, LMM**

Range: The permissible character set for the state descriptors are as follows:
 Alphabets, A-Z (upper case only)
 Numerals 0-9
 Underscore (_)

FL(1)–(1024) (Node Specif.)

Type: **Logical** **Box Flag Variables**—Each LMM has a set of 2048 local flag variables that can be used by process modules in this LMM to implement batch operations. These flags are local to the LMM and are different than the 12 logic-slot flags, and the 128 flags provided in each process module.

Lock: **Oper**

Default: **Off**

PtRes: **LMM**

Range: **Off** (Flag is off)
On (Flag is set)

Helpful Hint: For the first 128 flags, the On state is alarmed.

FLLSBA (Node Specif.)

Type: **Integer** **Least Significant Source Address**— Defines the least significant source address of a contiguous block of LC control relays (flags). When the value is -1, there are no flags.

Lock: **PtBld**

Default: **-1**

PtRes: **LMM**

Range: **-1 to 4095**

-G-**GENDESC (Logic)**

Type: **String_8** **Generic Descriptors**—Defines up to 12 generic descriptors that can be
Lock: **PtBld** assigned up to 12 logic channels. Refer to the description of the PRMDESC
Default: **Blank** parameter, and to the *LM Control Functions* manual for a detailed description.
PtRes: **NIM**
Range: Permissible character set for the eight-character generic descriptors consist of all characters on the
 Engineer' s Keyboard. Basically this set consists of alphabetics A-Z, numerics 0-9, and the following
 special characters: space ! " % & ' () * + - / : ; > < = ? _ , . \$

<i>Helpful Hint:</i> Example: GENDESC(7) is the descriptor for parameter PRMDESC(7), etc.

-H-

HIGHAL (AnalogIn, Digin, DigComp, Flag)

Type: **E:ALMTYPE** **Highest Alarm Detected**—This parameter is used by the system to ensure that when two or more different types of alarms occur on a point at the same time, the most important or highest level alarm appears on the point's Group, Detail, and Alarm Summary displays.

Lock: **View**

Default: **NoAlarm**

PtRes: **NIM**

Range: Analog Input Points:

NoAlarm (No alarm exists—lowest level alarm)

PVHI (PV High)

PVLO (PV Low)

BadPV (Bad PV—highest level alarm)

Digital Composite, Digital Input, and Flag Points:

NoAlarm (No alarm has been detected)

OffNorm (Current PV state is not the configured PVNORMAL state. For a flag point, the off-normal state (STATE1) is the alarmed state.)

UnCEvt (Uncertain event was detected. Does not apply to a flag point.)

CmdDis (Command Disagree; field device did not respond to commanded output state. Does not apply to a flag point.)

BadCtl (Bad control alarm, BADCTLFL = On. Digital Composite only.)

HIGHALPR (AnalogIn, DigComp, Digin, Flag)

Type: **E:ALPRIOR** **Highest Level Alarm's Priority**—Defines the priority of the highest alarm currently detected at the data point.

Lock: **View**

Default: **NoAction**

PtRes: **NIM**

Range: **JnlPrint** (Alarm is historized and reported to the printer, but not annunciated.) **Printer**

(Alarm is reported to the printer, but not historized and not annunciated.) **Emergency**

(Alarm is historized, annunciated, and reported to all alarm summary displays.)

High (Alarm is historized, reported to Area Alarm Summary Display and Unit Alarm Summary Display.)

Low (Alarm is historized, reported to the Unit Alarm Summary Display, and annunciated.)

Journal (Alarm is historized, but not reported to Universal Stations and not annunciated.)

NoAction (Alarm is not reported to the system and not annunciated.)

**I0-I2 (DigComp)**

Type: **Logical** **Override Interlocks for Output States 0 – 2**—Override interlocks are controlled by the LC. The operator and user program cannot change the output state when any override interlock is On. An override interlock is provided for each of the three states. Refer to the *LM Control Functions* manual for a detailed description.

Lock: **View**

Default: **Off**

PtRes: **LMM**

Range: **Off** (Override interlock has no effect on the point state)
On (Override interlock prevents UCN-initiated writes to OP)

Helpful Hint: The override function must be in ladder logic. Digital Composite points will update the OP in regard to the output states. This, in turn, reflects the override function.

INITMAN (DigComp)

Type: **Boolean** **Initialization Manual Flag**—Indicates when this point is in the Initialization Manual state.

Lock: **View**

Default: **Off**

PtRes: **LMM**

Range: **Off** (Not in Initialization Manual state)
On (In Initialization Manual state)

INPTDIR (AnalogIn)

Type: **E:POLARITY** **Analog Input Direction**—Specifies direct or reverse action for the PV input at this data point. Direct means PVCALC increases as PVRAW increases.

Lock: **Eng/PB**

Default: **Direct** Reverse means PVCALC decreases as PVRAW increases.

PtRes: **LMM**

Range: **0-Direct** (Highest energy from sensor = 100% PV)
1-Reverse (Highest energy from sensor = 0% PV)

Helpful Hint: INPTDIR configuration applies only if PVCHAR = Linear or Sqrroot.

INPTDIR (DigIn)

Type: **E:POLARITY** **Digital Input Direction**—Defines the contact condition required to light the upper or lower box on a Group or Detail Display for a digital input point. See Figure I-1.

Lock: **Eng/PB**

Default: **Direct**

PtRes: **LMM**

Range: **0-Direct** (Open $\geq$ PVRAW = Off; Closed $\leq$ PVRAW = On)
1-Reverse (Open $\geq$ PVRAW = On; Closed $\leq$ PVRAW = Off)

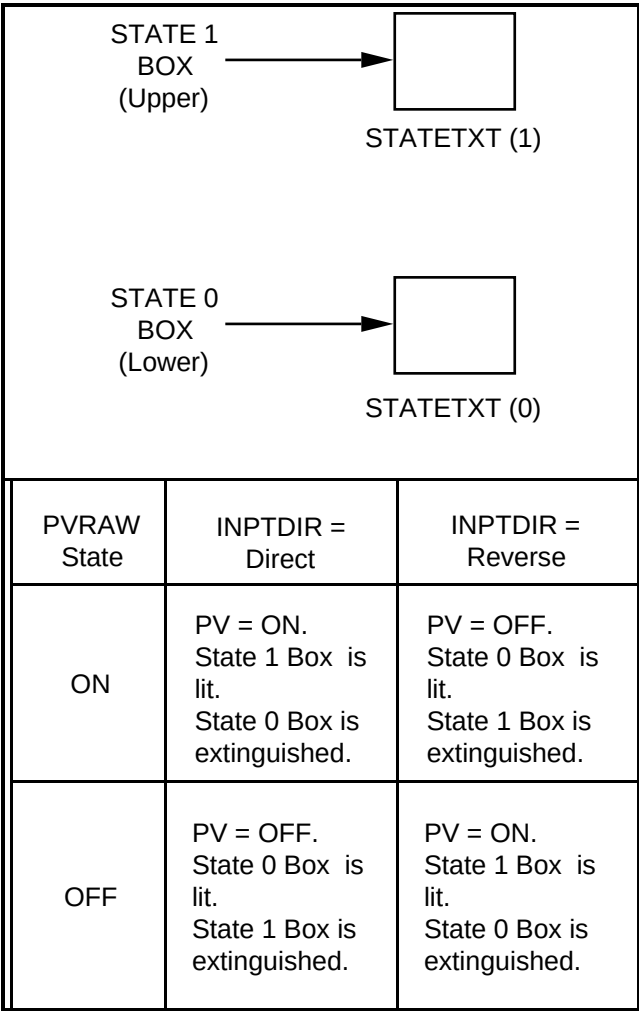


Figure I-1 — INPTDIR vs. Display Indicators

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-K-**KEYWORD (Analg I/O, Dig I/O, DigComp, Flag, Numeric, Timer)**

Type: **String_8** **Keyword Descriptor**— An eight-character descriptor that is used to describe an important aspect of this particular data point. For example, in Figure N-1 (see

Lock: **PtBld**

Default: **Blank** NAME) the keyword for the data point is REFLUX.

PtRes: **NIM**

Range: Alphabetics A-Z (upper case only).
 Numerics 0-9 (an all numeric keyword is not allowed).
 Underscore () cannot be used as the first character or the last character in a keyword.
 Consecutive underscores are not allowed.

-L-

L (Logic)

Type: **Real or Logical** **External Output Value**—Each input may be logical or real. The actual data type cannot be checked against the connection requirement during point building. A mismatch cannot be determined until run time; therefore, the user cannot be flagged regarding an error. Array (1..12).

Lock: **View**

Default: **NaN**

PtRes: **LMM**

Range: **N/A**

Helpful Hint: L, if accessed from the LCN, must be accessed as a logical data type.

LASTPV (AnalogIn)

Type: **Real** **Last PV Value**—Indicates the value of the PV before the value became bad (BADPVFL).

Lock: **View**

Default: **NaN**

PtRes: **LMM**

Range: **N/A**

LIBADOPT (Logic)

Type: **E:\$LIBADOP** **Logic Bad-Input Handling Option**—If a Boolean input is not successfully fetched for an input connection to the logic slot, its value is defaulted to one of the values (Off, On, Hold) selected through this parameter. Refer to the *LM Control Functions* manual for a detailed description.

Lock: **PtBld**

Default: **Hold**

PtRes: **LMM**

Range: **0-On** (On state is substituted for bad input)
1-Off (Off state is substituted for bad input)
2-Hold (Last good value is substituted for bad input)

LISRC(1)–(12) (Logic)

Type: **Universal** **Logic Input Connection Source**—Define the parameters whose current values are to be supplied to one or more of up to 12 logic slot inputs (L1 – L12). The parameters can be specified using the "Tagname.Parameter" format or the hardware reference address format. Refer to the *LM Control Functions manual* for a detailed description.

Lock: **PtBld**

Default: **Null**

PtRes: **LMM**

Range: Use Tagname.Parameter for tagged points where Tagname can be up to 8, or up to 16 characters (depending on the system option) and the permissible character set is as follows:
 Alphabetic A-Z (upper case only)
 Numerics 0-9 (an all numeric tag name is not allowed)
 Underscore (_) cannot be used as the first character or the last character, and consecutive underscores are not allowed.
 Embedded space characters are not allowed.
 An * is used to default to this point's tag name.
 Parameter name can be up to eight characters and must be a legitimate parameter name.
 Any boolean or real number that is compatible with the destination point (LODSTN(N)).

Some possible input-connection sources are:

- a. "DigIn slot Tagname.PVFL"
- b. "Node Specific Flag slot Tagname.PVFL(nnnn)" where nnnn = 1–1024
- c. "!Box.FL(nnnn)" for a node specific flag that resides in the same LM where nnnn = 1–1024
- d. "\$NMhhBxx.FL(nnnn)" for a node specific flag that resides in a different LM on the same UCN; hh is the NIM UCN address, xx is the LM's UCN node number, and nnnn = 1–1024
- e. "AnalgIn_Tag.PV"
 !BOX.NN(nnnn)
 !BOX.FL(nnnn)
 \$NMhhBxx.NN(nnnn)
 \$NMhhBxx.FL(nnnn)

Use the hardware reference address !ADRXXXX for direct connection to the LC.

LMNAME (Node Specif.)

Type: **String_8** **Names for a Redundant Pair of LMs**—A two-element array of up to eight characters per element, which names a redundant pair of LMs. Note that the LMs have no fixed position from which they can be identified.

Lock: **PtBld**

Default: **Pref**

PtRes: **Non-Pref**

Range: **Preferred**
Non-Pref

LMSRC (DigComp)

Type: **Integer** **Local Manual Source Address in the LC**

Lock: **PtBld**

Default: **-1**

PtRes: **LMM**

Range: **-1 to 8191**

LOADSCOP (UCN Node)

Type: **E:\$LOADSCP Load Scope**—Defines the scope of the point-build procedure for NIM and LM configuration. Indicates whether or not a given parameter is checked at the

Lock: **PtBld** NIM level or at the LM level. If at the NIM level, LM-resident parameters are

Default: **NIM Only** checked for form only, and not for validity.

PtRes: **NIM**

Range: **NimOnly** (Configured data is to be loaded in the NIM)
NimAnd LM (Configured data is to be loaded into the NIM and LM)

NOTE

When points are built to a NIM and the NIM is restarted with no database, the points need to be reloaded from checkpoint or the points must be reconfigured. If the database is to be reconfigured, the LMM must be in Idle, and the point execution state must be inactive. This allows the point build operation to override the database that already exists there.

NOTE

To delete entities from the LM database, the LOADSCOP parameter must be changed from NimAndLM and NimOnly. The database can then be reloaded, and the desired entities can be deleted.

LOCALMAN (DigComp)

Type: **Logical** **Local Manual Flag**—When On, indicates that the output(s) is being locally controlled. When this flag is on, it usually indicates that the "hand/off/auto" switch is not in the "auto" position. This flag is off when the Local Manual Source Address (LMSRC) equals -1. For more information, refer to subsection 3.3.13 in the *LM Control Functions* manual.

Lock: **View**

Default: **Off**

PtRes: **LMM**

Range: **Off**
On

LOCUTOFF (AnalogIn)

Type: **Real** **Low Signal Cut Off for Flow Inputs**—Defines the low cut-off point for flow inputs.

Lock: **Eng/PB**

Default: **NaN**

PtRes: **LMM**

Range: **PVEULO to PVEUHI**,
NaN

Helpful Hint: LOCUTOFF configuration applies only if PVCHAR = Linear or Sqrroot.

LODSTN(1)–(12)

LODSTN(1)–(12) (Logic)

Type: **Universal** **Logic Output-Connection Destination**—Specifies up to 12 destinations to which the current values of the logic slot outputs are supplied. The destinations can be specified using the "Tagname.Parameter" format or the hardware reference address format. Refer to the *LM Control Functions* manual for a detailed description.

Lock: **PtBld**

Default: **Null**

PtRes: **LMM**

Range: Use Tagname.Parameter for tagged points where Tagname can be up to 8, or up to 16 characters (depending on the system option) and the permissible character set is as follows:

Alphabetic A-Z (upper case only)

Numerics 0-9 (an all numeric tag name is not allowed)

Underscore (_) cannot be used as the first character or the last character, and consecutive underscores are not allowed.

Embedded space characters are not allowed.

An * is used to default to this point's tag name.

Parameter name can be up to eight characters, and must be a legitimate parameter name.

Some possible output-connection destinations are:

- a. "!Box.FL(nnnn)" for a node specific flag that resides in the same LM where nnnn = 1–1024.
- b. "\$NMhhBxx.FL(nnnn)" for a node specific flag that resides in a different LM on the same UCN; hh is the UCN number, xx is the LM node specific number of the destination parameter, and nnnn = 1–1024.

Use the hardware reference address !ADRXXX for connections directly to the LC.

LOENBL(1)–(12) (Logic)

Type: **Integer** **Logic Output Enable**—Allows the respective output connection defined by LODSTN to write the value of the specified logic slot parameter to the destination. The logic output is allowed when the enable function, selected from the list below, is On. -1 is always enabled.

Lock: **PtBld**

Default: **-1**

PtRes: **LMM**

Range: **-1 to 8191**

LOGMIX (Logic)

Type: **E:\$LOGMIX** **Logic Mix**—Defines the number of input and output connections that this logic slot is to contain.

Lock: **Pt Bld**

Default: **12_0_12**

PtRes: **LMM**

	<i>Input Connections</i>	<i>Number of</i>	<i>Output Connections</i>
<i>Range:</i>	<i>LISRC(1)-LISRC(12)</i>	<i>Logic Blocks</i>	<i>LOSRC(1)-LOSRC(12)</i>
12_0_12	12	0	12

LSPPXORN

LSPPXORN (Node Specif.)

Type: **Integer** **Last Hour's Point Processing Fetch/Store Overruns**—Indicates the number of point processing overruns that have been detected in the last hour.
Lock: **View**
Default: **0** LSPPXORN is set equal to the contents of parameter CRPPXORN, every hour
PtRes: **LMM** on the hour.
Range: **≥ 0**

LSUCNORN (Node Specif.)

Type: **Integer** **Last Hour's UCN Access Overruns**—LSUCNORN is set equal to the contents of parameter CRUCNORN, the current hour's UCN Access Overruns, every
Lock: **View** hour on the hour.
Default: **0**
PtRes: **LMM**
Range: **≥ 0**

-M-**MODATTR (AnalogOut, DigComp, DigOut)**

Type: E:MODATTR **Mode Attribute**—Defines whether an operator or sequence program has the authority to change certain parameters of this data point. At the Universal Station, the mode attribute is displayed next to the mode of the data point. If the mode attribute is Program, a -P appears to the left of MODE. If the attribute is operator, blanks are displayed to the left of mode.

Lock: Oper

Default: Operator

PtRes: LMM

Range: 0-None (No mode attribute)
 1-Operator (Operator can set Mode, OP, SP, Ratio, Bias)
 2-Program (Program can set Mode, OP, SP, Ratio, Bias)

Helpful Hint: MODATTR change requires REDTAG = Off. When the "normal mode" button on the Operator's keyboard is pressed, MODATTR = NMODATTR, unless NMODATTR = None.

MODE (AnalogOut)

Type: E:MODE **Mode**—Defines the current mode of the data point. Parameter MODATTR determines whether operator or the sequence program provides the output value for this point. If PNTFORM is Component, then MODE parameter is not applicable for this data point.

Lock: Oper

Default: Man

PtRes: LMM

Range: 0-None
 1-Man (Operator or Program provides the point's output value (OP)).
 2-Cas (Data point receives its output value from a primary data point. If RCASOPT is DDC, data point receives its output value from an AM point.)
 5-Normal (Parameter NMODE determines this point's mode.)

Helpful Hint: 1. MODE change by a program requires MODATTR = Program and REDTAG = Off.
 2. MODE change by an operator requires MODATTR = Operator, MODEPERM = Permit, and REDTAG = Off.

MODE (DigComp, DigOut)

Type: E:MODE **Mode** —Defines the current mode of the data point. Parameter MODATTR determines whether operator or the sequence program provides the output value for this point. If PNTFORM is Component and a DigOut point, then MODE parameter is not applicable for this data point.

Lock: Oper

Default: Man

PtRes: LMM

Range: 0-None
 1-Man (Operator or Program controls slot's output (OP)).
 5-Normal (Parameter NMODE contains slot's mode.)

Helpful Hint: 1. MODE change by a program requires MODATTR = Program, and REDTAG = Off.
 2. MODE change by an operator requires MODATTR = Operator, MODEPERM = Permit, and REDTAG = Off.

MODEAPPL (AnalgOut, DigComp, DigOut)

Type: **Logical, Array (1..4)**

Lock: **View**

Default: **Man= True**

Auto=False

Bcas=False

Cas=False

(True for AnalgOut)

PtRes: **NIM Mode Applicability**

Range: **N/A**

MODEPERM (AnalgOut, DigComp, DigOut)

Type: **E:MODEPERM Mode Permissive**—Determines whether the operator can change the mode

Lock: **Eng/PB** of this data point.

Default: **Permit**

PtRes: **LMM**

Range: **0-Permit** (Operator can change this point's mode)

1-NotPerm (Operator cannot change this point's mode)

MOMSTATE (DigComp)

Type: **E:\$MOMSTAT Momentary Output States**—Defines which of the output states are

Lock: **Eng/PB** momentary. Refer to the *LM Control Functions* manual for a detailed

Default: **None** description.

PtRes: **LMM**

Range: **0-None** (No momentary output states)

1-Mom_0 (State 0 is momentary if NOSTATES = 2)

2-Mom_1 (State 1 is momentary if NOSTATES = 2 or 3)

3-Mom_2 (State 2 is momentary if NOSTATES = 3)

4-Mom_1_2 (State 1 and State 2 are momentary; valid if NOSTATES = 3)

MOVPVFL (DigComp)

Type: **Logical Moving PV Flag**—Indicates whether the PV is moving from one state to

Lock: **View** another state.

Default: **Off**

PtRes: **LMM**

Range: **Off** (PV is not moving)

On (PV is moving)

MOVPVTEXT

MOVPVTEXT (Node Specif.)

Type: **String_8** **Moving PV Text Descriptor**—Defines the state descriptor that is displayed
Lock: **PtBld** when the digital composite point is changing states (moving from one state to
Default: **Moving** another), or is in-between states. This descriptor will be displayed for all digital
PtRes: **NIM** composite points in this LM.

Range: The permissible character set for the up-to-eight-character descriptor is as follows:
Alphabets A-Z (upper case only)
Numerics 0-9,
Underscore (_)

<i>Helpful Hint:</i> If NODENUM is an even number, this parameter is hidden on the UCN Node Configuration Display.
--

-N-**NAISLOT (Node Specif.)**

Type: **Integer** **Number of Analog Input Slots**—Refer to the *LM Control Functions* manual
Lock: **PtBld** for a detailed description of LM processing capacity.
Default: **0**
PtRes: **LMM**
Range: **0** to **127** for 0.5 second scan rate, 0 to 254 for 1 second scan rate

NAME

Type: **Ent_Id** **Tag Name**—Identifies this point to the system and on displays, reports, and
Lock: **Eng/PB** logs. Figure N-1 shows examples of the Group and Detail Displays on which
Default: **N/A** the tag name appears.
PtRes: **NIM**

Flag and numeric-type data points do not have to be configured by using the point builder (DEB). All other types of data points have to be configured by using the DEB and require that a tag name be specified during the point build process.

Range: Tag name can be up to 8, or up to 16 characters (depending on the system option), and the permissible character set is as follows:
 Alphabetics A-Z (upper case only)
 Numerics 0-9 (an all numeric tag name is not allowed)
 Underscore () cannot be used as the first character or the last character, and consecutive underscores are not allowed.
 Embedded space characters are not allowed.

NAOSLOT (Node Specif.)

Type: **Integer** **Number of Analog Output Slots**—Refer to the *LM Control Functions* manual
Lock: **PtBld** for a detailed description of LM processing capacity.
Default: **0**
PtRes: **LMM**
Range: **0** to **482** for 0.5 scan rate, 0-965 for 1 second scan rate.

NDCSLOT (Node Specif.)

Type: **Integer** **Number of Digital Composite Slots in a LM**—Refer to the *LM Control*
Lock: **PtBld** *Functions* manual for a detailed description of LM processing capacity.
Default: **0**
PtRes: **LMM**
Range: **0** to **304**

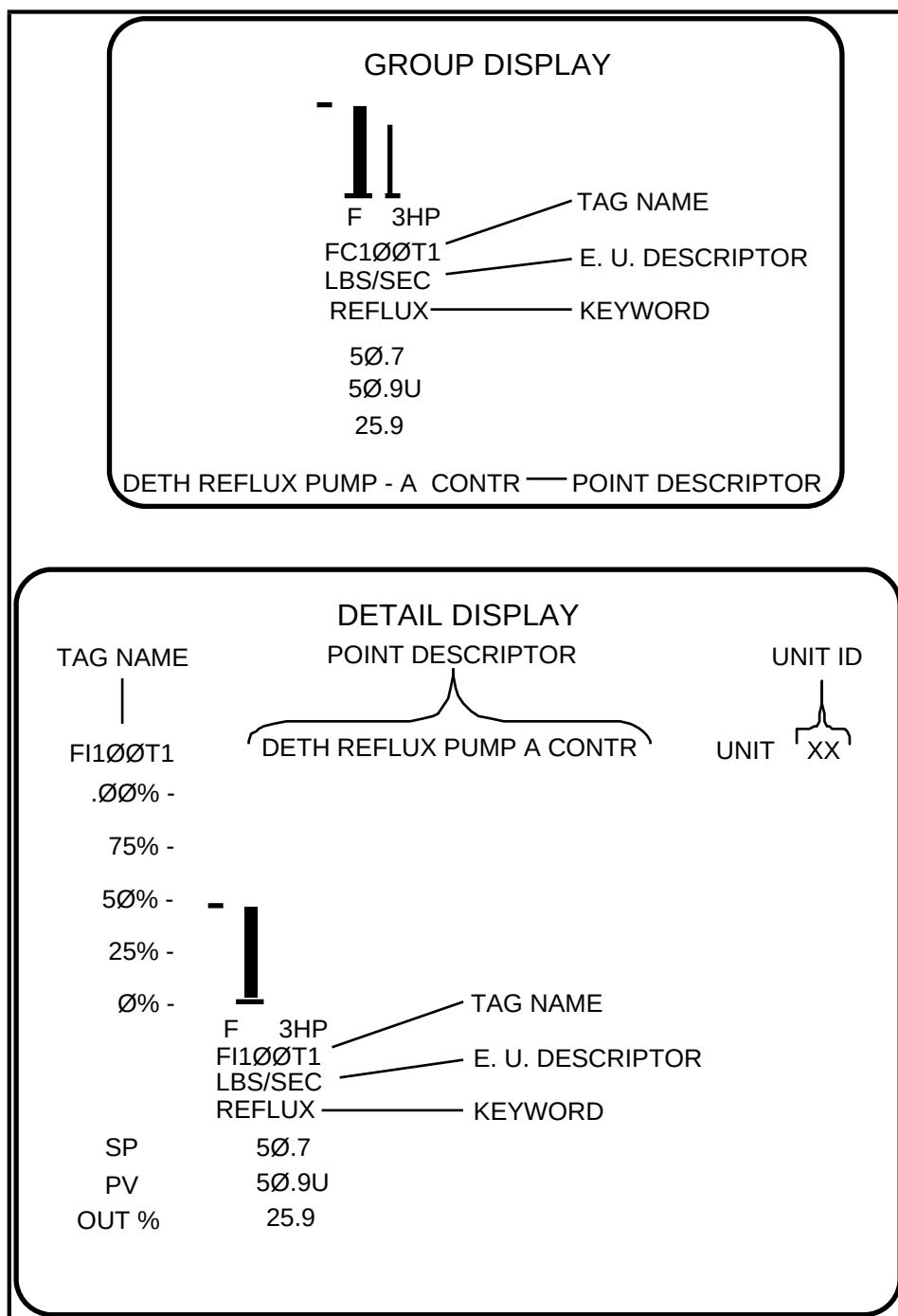


Figure N-1 — Locations of Terms on Group and Detail Displays

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NFLAG (Node Specif.)

Type: **Integer** **Number of Flags in the Node Specific Data Point.**
 Lock: **PtBld**
 Default: **0**
 PtRes: **LMM**
 Range: **0 to 1024**

NLOGSLOT (Node Specif.)

Type: **Integer** **Number of Logic Slots in the LM**—Refer to the *LM Control Functions* manual for a detailed description of LM processing capacity.
 Lock: **PtBld**
 Default: **0**
 PtRes: **LMM**
 Range: **0 to 14**

NMODATTR (AnalgOut, DigComp, DigOut)

Type: **E:MODATTR** **Normal Mode Attribute**—Defines whether an operator or a program can change certain parameters such as the mode, SP, or OP of a data point when the point is in the normal mode.
 Lock: **Engr**
 Default: **None**
 PtRes: **LMM**
 Range: **0-None** (MODATTR is not affected by this parameter)
 1-Operator (When the “Normal Mode” key is pressed, MODATTR changes to Operator)
 2-Program (When the “Normal Mode” key is pressed, MODATTR changes to Program)

Helpful Hint: If NMODATTR = Operator or Program and the "normal mode" key on the Operator's keyboard is pressed, MODATTR = NMODATTR. If NMODATTR is to be changed, the engineer must change it.

NMODE (AnalgOut)

Type: **E:MODE** **Normal Mode**—Allows an engineer to define the normal mode for this data point.
 Lock: **Engr**
 Default: **None**
 PtRes: **LMM**
 Range: **None** (No configured "normal" operating mode)
 Man (Manual is configured "normal" mode)
 Cas (Cascade is configured "normal" mode)

Helpful Hint: NMODE configuration for the Cas option requires RCASOPT = Ddc.

NMODE (DigComp, DigOut)

Type: **E:MODE** **Normal Mode**—Allows an engineer to define the normal mode for this data point.
 Lock: **View**
 Default: **None**
 PtRes: **LMM**
 Range: **1-Man** (Manual is configured "normal" mode)

NN(1)–(1024) (Node Specif.)

Type: **Real** **Numeric Value**—This is an array of 1024 numeric variables. The upper bound of this array is determined by parameter NNUMERIC.
Lock: **Oper**
Default: **NaN**
PtRes: **LMM**
Range: **NaN**
Real

NNLSBA (Node Specif.)

Type: **Integer** **Least Significant Source Address**—Defines the least significant source address of a contiguous block of LC internal registers (numerics). When the value is 0, it indicates that numerics are not in use.
Lock: **PtBld**
Default: **0**
PtRes: **LMM**
Range: **4096 to 8191**

NNUMERIC (Node Specif.)

Type: **Integer** **Number of Numerics in LM Node Specific Data Point**
Lock: **PtBld**
Default: **0**
PtRes: **LMM**
Range: **0 to 1024**

NTIMER (Node Specif.)

Type: **Integer** **Number of Timer Slots in LM Node Specific Data Point**
Lock: **PtBld**
Default: **0**
PtRes: **LMM**
Range: **0 to 700**

NOCINPTS (Node Specif.)

Type: **Integer** **Number of Control Input Slots**—Defines the number of digital input slots in this LM.
Lock: **PtBld**
Default: **0**
PtRes: **LMM**
Range: **0 to 1866**

NOCOPTS (Node Specif.)

Type: **Integer** **Number of Digital Output Slots**—Defines the number of digital output slots.
Lock: **PtBld**
Default: **0**
PtRes: **LMM**
Range: **0 to 4000**

NODEASSN

NODEASSN (UCN Node)

Type: E:\$NODEASN **Node Assignment**—Defines whether this NIM or another NIM is the
Lock: PtBld primary NIM for this UCN. When multiple NIM modules are connected to
Default: ThisNIM the UCN, it becomes the primary NIM for each node specific component if
PtRes: NIM selected.
Range: ThisNIM (This NIM is the primary NIM for the UCN. ThisNIM supports parameter fetch/store, alarms, AM control strategy and checkpointing.)

RemotNIM (Another NIM is the primary NIM for this UCN. The RemotNIM is connected to a separate LCN network. During node specific configuration, checkpointing and database restores occur on the RemotNIM.)

Helpful Hint: NODEASSN configuration requires NODETYP = NIM or LM.

NODENUM (Node Specif., UCN Node)

Type: Integer **Node Number**—Defines the address of the NIM and the LMs on the UCN.
Lock: PtBld NODENUM assigned for any LM must be odd whether PKGOPT equals
Default: N/A Option 1 (nonredundant) or Option 2 (redundant). Because of this restriction
PtRes: LMM and because the NIM takes up one odd address and the next even address, the
maximum number of LMs that can be on the UCN is 31. The primary LMM is
assigned an odd address, the associated secondary (redundant) LM is assigned
the next (even) address.

Range: 0 to 64

NOTE

The node number assigned to the NIM must be the lowest node number on the UCN.

NODENUM (Analg I/O, Dig I/O, DigComp, Flag, Logic,Timer, Numeric)

Type: Integer **Node Number**—Defines the UCN address of the LM in which this point
Lock: PtBld resides.
Default: None
PtRes: NIM
Range: 1 to 64

NODESC (Logic)

Type: **Integer** **Number of Generic Descriptors**—Defines the number of user-defined generic descriptors that are to be used on this logic slot. For each descriptor, the
Lock: **PB** parameter in the logic slot to which the generic descriptor is attached is defined
Default: **0** by the PRMDESC(n) parameter, and the corresponding descriptors are defined
PtRes: **NIM** by the GENDESC(n) parameters. This allows the user to customize the
 descriptors used for displaying the logic slot on the Universal Station displays.

Range: **0 to 12**

NODESTS (UCN Node)

Type: **E:\$DSPSTAT** **NIM Node Summary Status**—Indicates the current overall status of the
Lock: **View** NIM.
Default: **N/A**
PtRes: **NIM**
Range: **ConfigMis** (Configuration mismatch)
 Fail (Complete failure)
 NotSync (Not synchronized)
 OK (NIM is performing normally)
 PartFail (Partial failure)
 OffNet (NIM cannot communicate with LMM)

NODESTS (Node Specif.)

Type: **E:\$DSPSTAT** **LM Node Summary Status**—Indicates the current overall status of the LM
Lock: **View** on the UCN.
Default: **N/A**
PtRes: **LMM**
Range: **0-OK** (LM is performing normally)
 2-PF_IDLE (Partial failure in LMM that is in Idle state)
 4-PF_UmdlL (Partial failure in LMM that is in Idle state)
 5-PartFail (Partial failure in LMM that is in run state)
 6-Failed (LMM has sent a "failed" message to the NIM)*
 7-Alive (No event reports or point processing)*
 8-Loading (LMM's personality or database is loading)*
 9-PowerOn (Ac and dc power is on at this LMM)*
 10-OffNet (NIM cannot communicate with LMM)
 12-NotConfig (LMM cannot be found on the UCN)*
 14-UnAvail (Unavailable)
 15-UndTst (Under test)*

* Presently not used

Helpful Hint: Loading the LMM's operating personality requires NODESTS = Alive.

NODESTAT

NODESTAT (UCN Node)

Type: **E:\$NODESTA** **NIM's Node Status**
Lock: **View**
Default: **N/A**
PtRes: **NIM**
Range: **ConfigMis** (NIM has a configuration mismatch)
 OffNet (NIM is not configured on UCN)
 Run (NIM is configured and running)

NODETYP (UCN Node)

Type: **E:\$UCNNDTY** **UCN Node Type**—Defines the type of node on the UCN.
Lock: **PtBld**
Default: **NIM**
PtRes: **NIM**
Range: **NIM** (Network Interface Module)

NODETYP (Node Specif.)

Type: **E:\$UCNNDTY** **UCN Node Type** —Defines the type of node on the UCN.
Lock: **PtBld**
Default: **NotConfig**
PtRes: **NIM**
Range: **NotConfig** (Not Configured)
 LM (Logic Manager)

NODETYP (Analog I/O, Dig I/O, DigComp, Flag, Logic, Numeric, Timer)

Type: **E:\$UCNNDTY** **UCN Node Type** —Defines the type of node on the UCN.
Lock: **View**
Default: **NotConfig**
PtRes: **NIM**
Range: **LM** (Logic Manager)

NODFSTAT (Node Specif.)

Type: **E:\$NODFSTA** **Node's Functional Status**—Defines the status of the UCN node.
Lock: **Supr**
Default: **Basic**
PtRes: **NIM**
Range: **Full** (All LCN devices can read/write from/to this node)
Basic (AM and CM cannot write to this node)

NODINPTS (DigComp)

Type: **Integer** **Number of Digital Inputs**—Defines the number of digital input connections to this data point.
Lock: **PtBld**
Default: **1**
PtRes: **LMM**
Range: 0- (No inputs)
1- (One input)
2- (Two inputs)

NODOPTS (DigComp)

Type: **Integer** **Number of Digital Outputs**—Defines the number of digital output connections from this data point.
Lock: **PtBld**
Default: **1**
PtRes: **LMM**
Range: 0- (No outputs)
1- (One output)
2- (Two outputs)
3- (Three outputs)

NOLINPTS (Logic)

Type: **Integer** **Number of Logic Input Points**—Indicates the number of inputs to this Logic Block.
Lock: **View**
Default: **N/A**
PtRes: **LMM**
Range: **12**

NOLOGBLK (Logic)

Type: **Integer** **Number of Logic Blocks**—Indicates the number of logic blocks that have been configured for a particular logic slot.
Lock: **View**
Default: **None**
PtRes: **LMM**
Range: **0**

<i>Helpful Hint:</i> In the LM Logic Block points are intended for data transfer only.
--

NOLOPTS

NOLOPTS (Logic)

Type:	Integer	Number of Logic Points	Indicates the number of outputs from this Logic
Lock:	View	Block.	
Default:	N/A		
PtRes:	LMM		
Range:	12		

NONECONF (DigComp)

<i>Type:</i>	Boolean	Add Optional None State —In addition to the standard three output states, [1=1st active state, 0=inactive (middle) state, and 3=2nd active state],
<i>Lock:</i>	PtBld	
<i>Default:</i>	Off	NONECONF = On adds state 2, the None state. If the output state read back
<i>PtRes:</i>	LMM	from the Logic Controller is as defined in NONE_OP1 through NONE_OP3, no
		alarm is generated, even though the state read back does not match the
		commanded state in OP.

Range: **Off** (None state not configured)
 On (None state configured)

NONE_OP(1)–(3)

Type:	Boolean	Value to be Stored in Output n When the Output is in the None State (n = output 1, 2, or 3.)
Lock:	EngPB	
Default:	Off	
PtRes:	LMM	
Range:	Off On	

Helpful Hint: NONE_OP1 is configured only if NODOPTS ≥ 1, NONE_OP2 is configured only if NODOPTS ≥ 2, and NONE_OP3 is configured only if NODOPTS = 3.

NONETXT (DigComp)

Type:	String_8	State Descriptor for None State —If the output state read back from the Logic
Lock:	PtBld	Controller is as defined in NONE_OP1 through NONE_OP3, the character
Default:	N/A	string configured in NONETXT is transferred into FINSTTXT(2), and thus
PtRes:	LMM	becomes that state shown for OP for this point on the Group and Detail displays.

Range: The permissible character set for the state descriptors are as follows:
 Alphabets, A-Z (upper case only)
 Numerals 0-9
 Underscore (_)

NONSTATES

NOSTATES (DigComp)

Type: **Integer** **Number of Digital States**—Defines the number of states in this point.
Lock: **PtBld**

- State 1 is the first active state

Default: **2**

- State 0 is the inactive (middle) state

PtRes: **LMM**

- State 2 is the second active state

Refer to the *LM Control Functions* manual for a detailed description.

Range: 2-(two states can be configured: STATE 0 and STATE 1)
3-(three states can be configured: STATE 0, STATE 1, and STATE 2)

NTWKNUM (Analg I/O, Dig I/O, DigComp, Timer, Flag, Logic, Numeric, Node Specif., UCN Node)

Type: **Integer** **Network Number**—Defines the UCN network on which the NIM and LMs
Lock: **PtBld** reside.
Default: **N/A`**
PtRes: **NIM**
Range: **1 to 20**

-O-

OFFNRMFL (DigIn, DigComp)

Type: **Logical** **Off-Normal Alarm Flag**—Indicates whether an off-normal alarm has been
Lock: **View** detected at this data point.
Default: **Off**
PtRes: **LMM**
Range: **Off** (No alarm)
 On (Current PV state is not the same as the configured PVNORMAL state.)

OFFNRMPR

Type: **E:ALPRIOR** **Offnormal Alarm Priority**—Indicates priority of the off normal or change of
Lock: **Eng** state alarms.
Default: **NoAction** With R670, the default value is changed to NoAction.
PtRes: **NIM**
Range: **JnlPrint** (Alarm is historized and reported to the printer, but not annunciated.) **Printer**
 (Alarm is reported to the printer, but not historized and not annunciated.) **Emergency**
 (Alarm is historized, annunciated, and reported to all alarm summary displays.)
 High (Alarm is historized, reported to Area Alarm Summary Display and Unit Alarm Summary Display.)
 Low (Alarm is historized, reported to the Unit Alarm Summary Display, and annunciated.)
 Journal (Alarm is historized, but not reported to Universal Stations and not annunciated.)
 NoAction (Alarm is not reported to the system and not annunciated.)

OISRC(1)–(3) (DigComp)

Type: **Integer** **Override Interlock Source Addresses in the LMM**
Lock: **PtBld**
Default: **-1**
PtRes: **LMM**
Range: **-1 to 8191**

OP (AnalgOut)

Type: **Real** **Output in Percent**—Defines the output value from this point in percent.
Lock: **Oper**
Default: **0% of full**
 scale
PtRes: **LMM**
Range: **0 to 100%**

Helpful Hint: To manually change the output value requires MODE = Man and REDTAG = Off.

OP (DigComp)

Type: E:STATETXT **Last Digital Output State Requested**—The last digital state commanded by an operator at a Universal Station or by a user-written program. User programs and schematics (custom displays) write the commanded state into OP and the new value is copied into OPFINAL. Commands from the Operator Keyboard are written into the OPFINAL and copied into OP.

Lock: Oper

Default: STATETXT(0)

PtRes: LMM

Range: STATETXT(0) Descriptor (state 0, middle or inactive)
 STATETXT(1) Descriptor (state 1, Active)
 STATETXT(2) Descriptor (state 2, None)
 STATETXT(3) Descriptor (state 3, Active)
 If the state is equal to or greater than 4, OP contains Undefined.

Helpful Hint: OP contains the text that represents the last-commanded output state (for example: FORWARD, STOP, or REVERSE). A change in the value in OP requires MODE = Man and REDTAG = Off.

OP (DigOut)

Type: E:STATEXT **Digital Output**—The OP parameter is a self-defining enumeration of STATETXT and reflects the status of the SO parameter.

Lock: Oper or Um_Cont_Ctrl

Default: STATEXT(0)

PtRes: LMM

Range: STATETXT(0) Descriptor
 STATETXT(1) Descriptor

OPFINAL (AnalogOut)

Type: Real **Percent Output at the Control Element**—Output value after direct or reverse control action and output characterization have all been applied. If output has been configured for direct action (OPTDIR), 0.0% represents 0 LC value to the control element and 100% represents 4095 LC value. If configured for reverse action, 0.0% represents 4095 LC value, and 100% represents 0 LC value.

Lock: View

Default: 0% of full scale

PtRes: LMM

Range: 0 to 100%

OPFINAL (DigComp)

Type: **E:FINSTTXT** **Final Output State Read Back From the Logic Controller**—Override interlocks from the Logic Controller and the ladder logic program can alter the actual output(s) to the process so that it differs from the commanded state, the LC final output state is read back into OPFINAL [where more than one LC output for this point is in use, the LMM compares each output with OP, and reports the first match in OPFINAL. A Bad Control alarm (BADCTFL = On) is generated if no match is found on any output.] User programs and schematics (custom displays) write the new commanded state into OP and the new value is copied into OPFINAL and OP. When a new state command is issued from the LCN, OPFINAL receives the new state, but during the next LC scan, that state is replaced by the final output value read back from the LC.

Range: **FINSTTXT (0)** Descriptor (state 0, middle or inactive)
FINSTTXT (1) Descriptor (state 1, Active)
FINSTTXT (2) Descriptor (state 2, None)
FINSTTXT (3) Descriptor (state 3, Active)
 If the state is equal to or greater than 4, OP contains Undefined.

OPFINHI (AnalogOut)

Type: **Floating** **Output High Value at the Control Element**—Indicates the high value for raw analog output data.
Lock: **Engr**
Default: **4095**
PtRes: **LMM**
Range: **Greater than OPFINLO + 1** (If the output data exceeds this value, it is clamped at the OPFINHI value)

OPFINLO (AnalogOut)

Type: **Floating** **Output Low Value at the Control Element**—Indicates the low value for raw analog output data.
Lock: **Engr**
Default: **0**
PtRes: **LMM**
Range: **Less than OPFINHI - 1** (If the output is lower than this value, it is set to the OPFINLO value)

OPRATRFL (DigComp)

Type: **Logical** **Operator Mode Attribute Flag**—Indicates whether the current mode attribute is Operator.
Lock: **View**
Default: **None**
PtRes: **LMM**
Range: **Off** (Current mode attribute is Program or None)
On (Current mode attribute is Operator)

OPSTTEXT (DigComp)

Type: **E:STATETXT** **Output State Text**—Contains the state descriptor configured in STATETXT(0) through STATETXT(2) that represents the last-commanded output state value in OP. The value in OPSTTEXT is used to light the commanded state box that represents the last-commanded state on the Group and Detail display for this point. Because the commanded state can differ from the final output state from the Logic Controller, the text in OPSTTEXT can differ from the text in FINSTTXT.

Lock: **View**

Default: **N/A**

PtRes: **NIM**

Range: The permissible character set for the state descriptors are as follows:
 Alphabetics, A-Z (upper case only)
 Numerals 0-9
 Underscore (_)

OROPT (DigComp)

Type: **Logical** **Override Option**—Defines whether the operator can put the point in a bypass state where any of the I0-I2 inputs can be bypassed. Refer to the *LM Control Functions* manual for a detailed description.

Lock: **Eng/PB**

Default: **Off**

PtRes: **LMM**

Range: **Off** (No override)
On (Inputs can be overridden)

OPTDIR (AnalogOut)

Type: **E:POLARITY** **Analog Output Direct/Reverse Action**—Defines the output action of the OPFINAL value of the data point.

Lock: **Eng/PB**

Default: **Direct**

PtRes: **LMM**

Range: **Direct** (For final OP: 0% = LC value; 100% = 4095 LC value)
Reverse (For final OP: 0% = 4095 LC value; 100% = 0 LC value)

OVERVAL

OVERVAL(AnalIn)

Type: Integer **Overview Value in Percent**—Defines the amount of deviation (PV - SP, in percent) that causes the PV to reach the overview limit. As shown in Figure O-1, the baseline shows the normal operating value for this PV.

Lock: Eng/PB

Default: 25

PtRes: NIM

Range: 0 to 100 (Entering a 0 suppresses the value; value is not shown on the display)

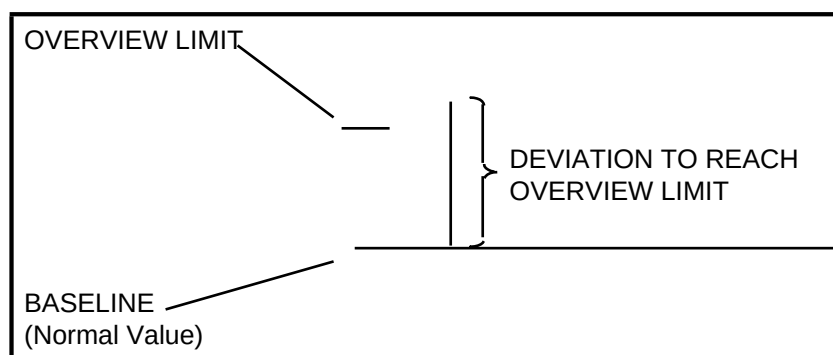


Figure O-1 — Overview Limit

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OUTIND (AnalOut)

Type: E:POLARITY **Reverse Output Indicator**—This R620 parameter defines how the output of regulatory data point is to be displayed by the Universal Station.

Lock: DEB for AO

Default: Direct

PtRes: NIM

Range: **Direct** (A 4 mA output from this data point is displayed as 0% at the Universal Station. A 20 mA output is displayed as 100%.)
Reverse (A 20 mA output from this data point is displayed as 0%. A 4 mA output is displayed as 100%.)

-P-

P0-P2 (DigComp)

Type: Logical **Permissive Interlocks for Output States 0, 1, & 2**—Permissive interlocks determine whether the operator and user program are allowed to use the respective state, or are locked out from that state. A permissive interlock (P0 – P2) is provided for each state (STATE0 – STATE2). The permissive interlocks themselves never cause the outputs to change. Refer to the *LM Control Functions* manual for a detailed description.

Lock: View

Default: On

PtRes: LMM

Range: Off (Respective state is locked out)
On (Respective state is permitted to be used)

Helpful Hint: P0 – P2 configuration requires PTEXECST = InActive or PNTSTATE = Idle.

PISRC(1)–(3) (DigComp)

Type: Integer **Permissive Interlock (P0 – P2) connection in the LC.**

Lock: PtBld

Default: -1

PtRes: LMM

Range: -1 to 8191

Helpful Hint: When unconfigured (-1), may be written by another UCN device.

PLCADDR (AnalogIn, AnalogOut, Flag, Numeric, Timer)

Type: Integer **Target Address in the LC for the Timer Run Contact.**

Lock: PtBld

Default: See Chart
Below

PtRes: LMM

Range: See Chart Below

Point Type	Default	Range	Lock
Analog Input	-1 (NotConf)	-1 to 8191	PtBld
Analog Output	-1 (NotConf)	-1 to 2047, 4097 to 8191	PtBld
Digital Input	-1 (NotConf)	-1 to 8191	PtBld
Digital Output	-1 (NotConf)	-1 to 8191	PtBld
Flag	-1, 0	-1 to 4095	View
Numeric	-1, None	-1, 4096 to 8191	View
Timer	-1 (NotConf)	-1 to 4095	PtBld

PMMCMD (Node Specif.)

Type: **E:\$PMMCMD** **LMM Command**
Lock: **OnProc**
Default: **None**
PtRes: **LMM**

NOTE

When points are built to a NIM and the NIM is restarted with no database, the points need to be reloaded from checkpoint or the points must be reconfigured. If the database is to be reconfigured, the LMM must be in Idle, and the point execution state must be Inactive. This allows the point build operation to override the database that already exists there.

Range: **0-None** (No command request has been issued)
1-Run (To "Run" state for processing points)
2-Idle (To "Idle" state for reloading the database)
4-ShutDown (To "Alive" state for reloading personality)
5-RsUcnLsb (Reset the Local Statistics Block to zeroes)
10-SwapPri (Switchover to the redundant LM)

PMMCMDTL (Node Specif.)

Type: **Blind Record** **LMM Command Processor Detail Record**
Lock: **View**
Default: **None**
PtRes: **LMM**
Range: **None** (No command request has been issued)
Null

PMMFLPOS (Node Specif.)

Type: **E:\$POSITIN** **Preference**—Preference refers to which LMM takes control in a contention
Lock: **View** situation, and to how the LMM's default responds to the odd/even UCN
Default: **N/A** address.
PtRes: **LMM**
Range: **Preferred**
Non-Preferred

PMMOPER (Node Specif.)

Type: **E:\$OPERAT** **Logic Manager Module Operation**—Indicates redundancy.
E
Lock: **View**
Default: **N/A**
PtRes: **LMM**
Range: **NonRedun** (This LMM has no redundant LMM to back it up)
Primary (This LMM is the primary point processor)
Secondry (This LMM is the secondary LMM that backs up the primary LMM)

PMMSFST (Node Specif.)

Type: **Blind Record** **Current LMM softfail status**— Defines the current LM softfail status.
Lock: **View** Blind record.
Default: **N/A**
PtRes: **LMM**
Range: **N/A**

PMMSTS (Node Specif.)

Type: **E:PMMSTS** **LMM Primary Status**
Lock: **View**
Default: **N/A**
PtRes: **LMM**
Range: **0-OffNet** (NIM cannot communicate with the LMM)
1-ConfgMs (Configuration mismatch detected)
2-Idle (Event reports but no point processing)
3-IdleSF (Soft failure occurred in Idle state)
4-OK (LMM is operating normally)
5-SoftFail (Soft failure while LMM is running)
6-Fail (LMM can be accessed but CPU is halted; box hard failure has been detected)
10-Alive (No event reports or point processing)
11-AliveSF (Soft failure during Idle state)
12-Test (LMM is in the test mode)
13-TestSF (Soft failure has been detected while the LMM is in the test mode)
15-Loading (Personality or database is loading)

Helpful Hint: Loading the LMM's operating personality software requires PMMSTS = Alive.
 Loading the LMM's database requires PMMSTS = Idle. Use parameter PMMCMD's "Shutdown" and "Idle" command requests, respectively.

PNTFORM (Analg I/O, Dig I/O, DigComp, Flag, Logic, Numeric, Timer)

Type: **E:\$PNTFORM** **Point Form**—Defines the form of the data point that is implemented.
Lock: **PtBld** Refer to the *LM Control Functions* manual for a detailed description of this
Default: **Full** function.
PtRes: **LMM**
Range: **Full** (Point is fully displayed and alarmed)
Componnt (Point is partially displayed but not alarmed)

PNTSTATE (Analg I/O, Dig I/O, DigComp, Flag, Logic, Timer)

Type: **E:PNTSTATE** **Point's Overall State**—Defines the state of the data point, which is based on
Lock: **View** the state of the LMM.
Default: **None**
PtRes: **NIM**
Range: **Config** (Configuration problem, i.e., mismatch, etc.)
Fail (NIM cannot communicate with LMM)
Idle (Point's LMM is in the Idle State)
OK (Point's LMM is the Run State and is OK)
Test (Point's LMM is being tested)
Saved [A SAVE DATA (checkpoint) operation from this point's LMM is in progress.]

PNTTYPE

PNTTYPE (AnalogIn, AnalogOut, DigComp, Flag, Logic, Numeric, Timer)

Type: **E:PNTTYPE** **Point Types**—Defines the type of point in the LM. Writing this parameter causes point initialization. Although “not configured” is in the range of values, it is never written to the system. Instead, it triggers a point deletion and reinitialization, and only the actual point type is stored.

Lock: **PtBld**

Default: **See Note Below**

PtRes: **LMM**

Range: **See Chart Below**

Point Type	Range
Analog Input	AnalogIn
Analog Output	AnalogOut
Digital Input	DigIn
Digital Output	DigOut
Digital Composite	DigComp
Flag	Flag
Logic	Logic
Numeric	Numeric
Timer	Timer

NOTE: Values are determined from the NIM Build Type Select menu.
These values are not configured.

PRIMMOD (AnalogIn, DigIn, DigComp)

Type: **Ent_Id** **Primary Module Identifier**—This parameter contains the tag name of the PM process module point to which this data point is assigned. The name of the Batch History Data Point that is used by the AM for collecting batch histories is specified by this tag name, and collects batch history for this data point.

Lock: **PtBld**

Default: **Null**

PtRes: **NIM**

Range: Tag name of the process module point can be up to 8 or up to 16 characters (depending on the system option) and the permissible character set is as follows:

- Alphabetics A-Z (upper case only)
- Numerics 0-9 (an all numeric tag name is not allowed)
- Underscore (_) cannot be used as the first character or the last character, and consecutive underscores are not allowed.
- Embedded space characters are not allowed.

PTDESC (Analg I/O, Dig I/O, Numeric, Logic)

Type: **String_24** **Point Descriptor**—A 24-character descriptor which is used to describe the point and appears on the Group and Detail Displays for the point. Refer to Figure N-1.

Lock: **PtBld**

Default: **Blank**

PtRes: **NIM**

Range: Permissible character set consists of all characters on the Engineer's Keyboard. This set consists of alphabetics A-Z, numerics 0-9, and the following special characters: space ! " % & ' () * + - / : ; > < = ? _ , . \$

PTEXECST (Analg I/O, Dig I/O, DigComp, Logic, Timer)

Type: **E:PTEXECST** **Point Execution State**—Defines the current execution state of the point.

Lock: **Supr**

Default: **Inactive**

PtRes: **LMM**

Range: **Inactive** (Point is not scanned or processed)
Active (Point is scanned and processed)

PTINAL (AnalgIn, DigComp)

Type: **Logical** **Point in Alarm Indicator**—Indicates when an alarm condition has been detected at this point. Set to ON (True) when any point alarm condition exists.

Lock: **View**

Default: **Off**

PtRes: **LMM**

Range: **Off** (Point is not in alarm)
On (Point is in alarm)

PV (AnalgIn)

Type: **Real** **Process Variable**—PV is the PV's current value after the PV is selected from one of the following possible sources: a field device, an operator, or a program. See PVSRCOPT and PVSOURCE.

Lock: **Oper**

Default: **NaN**

PtRes: **LMM**

Range: **PVEXEUIH** to **PVEXEULO**, **NAN**

Helpful Hint: PV change by a program requires PVSRCOPT = All and PVSOURCE = Sub.
PV change by an operator requires PVSRCOPT = All and PVSOURCE = Man.

PV (DigComp)

Type: **E:PVSTATES** **Current State**—PV is the PV's current state after the PV is selected from one of the following possible sources: a field device, an operator, a program, or the output of the PV's data point. See PVSRCOPT and PVSOURCE.

Lock: **Oper**

Default: **Bad**

PtRes: **LMM**

Range: **PVSTATES (0)** (Defined by STATETXT (0))
PVSTATES (1) (Defined by STATETXT (1))
PVSTATES (2) (Defined by BADPVTXT)
PVSTATES (3) (Defined by MOVPVTXT)
PVSTATES (4) (Defined by STATETXT(2))

Helpful Hint: PV change by a program requires PVSRCOPT = All and PVSOURCE = Sub.
PV change by an operator requires PVSRCOPT = All and PVSOURCE = Man.

PV (DigIn)

Type: **E:STATETXT** **Current State**—Indicates the current state of the PV input to this data point.
 Lock: **Oper**
 Default: **Off**
 PtRes: **LMM**
 Range: **STATETXT(0) or STATETXT(1)**

Helpful Hint: PV change by a program requires PVSRCOPT = All, and PVSOURCE = Sub.
 PV change by an operator requires PVSRCOPT = All, and PVSOURCE = Man.

PV (Flag)

Type: **E:STATETXT** **Current State**—Indicates the current state of the flag data point, and it is derived from PVFL. STATETXT(1) is the alarmed state.
 Lock: **Oper**
 Default: **Off**
 PtRes: **LMM**
 Range: **STATETXT(0) or STATETXT(1)**

PV (Numeric)

Type: **Real** **Process Variable**—Indicates the value of the numeric. This value maps into parameter NN(n) in the LM box where n = SLOTNUM.
 Lock: **Oper**
 Default: **NaN**
 PtRes: **LMM**
 Range: **-65535 to 65535**

PV (Timer)

Type: **Real** **Current Value**—Indicates the current time in seconds or minutes. The timer starts at 0 and is incremented towards the preset time established by the SP parameter.
 Lock: **View**
 Default: **0**
 PtRes: **LMM**
 Range: **0 to 65535**

PVALDB (AnalogIn)

Type: **E:PVALDB** **PV Alarm Deadband as a Percent of Full Range**—Alarm deadband is used to prevent excessive recurrence of alarms by adjusting the percent of Engineering Unit range at which the alarm "returns to normal." An increase in the deadband percent of PV range (PVEULO to PVEUHI) causes PV or deviation alarms to return to normal less frequently.
 Lock: **Eng/PB**
 Default: **One**
 PtRes: **LMM**

Range: **0-Half** (1/2 of 1% of Engineering Unit range)
1-One (1% of Engineering Unit range)
2-Two (2% of Engineering Unit range)
3-Three (3% of Engineering Unit range)
4-Four (4% of Engineering Unit range)
5-Five (5% of Engineering Unit range)
6-Six (6% of Engineering Unit range)

PVALDBEU (AnalogIn)

Type: **E:PVALDB** **PV Alarm Deadband in Engineering Units**—Alarm deadband is used to
Lock: **Eng/PB** prevent excessive recurrence of alarms by adjusting the percent of Engineering
Default: **1.0** Unit range at which the alarm "returns to normal."
PtRes: **LMM**
Range: **0.0 or greater**

PVAUTO (AnalogIn)

Type: **Real** **PV Auto Value**—Value of the PV after PVCALC is range checked, filtered,
Lock: **View** and clamped.
Default: **NaN**
PtRes: **LMM**
Range: **N/A**

PVAUTO (DigComp)

Type: **E:\$PVSTATS** **Current PV State**—Indicates the current PV state, based on the states of the
Lock: **View** inputs to the point.
Default: **BADPVTXT**
PtRes: **LMM**
Range: **STATETXT(0)**
STATETXT(1)
BADPVTXT
MOVPVTXT
STATETXT(2) (only if NOSTATES is 3)

PVAUTO (DigIn)

Type: **E:STATETXT** **Current PV State** —Indicates the current PV state that corresponds to the
Lock: **View** field contact input after direct/reverse correction.
Default: **None**
PtRes: **LMM**
Range: **STATETXT(0) or STATETXT(1)**

PVAUTOST (AnalogIn)

Type: **E:PVVALST** **PV Auto Value Status**—Indicates the current status of the PVAUTO value.
Lock: **View**
Default: **Bad**
PtRes: **LMM**
Range: **0-Bad** (All inputs, or result in PVCALC is bad)
1-Uncertn (Final result in PVCALC is an uncertain value)
2-Normal (Final result in PVCALC is a normal value)

PVCALC (AnalogIn)

Type: **Real** **Calculated PV**—PVCALC is the PV value in Engineering Units after the raw PV (PVRAW) input to this data point has been characterized.
Lock: **View**
Default: **NaN**
PtRes: **LMM**
Range: **PVEXEUHI to PVEXEULO, NaN**

PVCHAR (AnalogIn)

Type: **E:VALCHAR** **PV Characterization Option**—Defines the display characterization to be used for characterizing the input PV value. Characterization is based on the field sensor type. Sensor type is selected at the LC Analog Input card. When PVCHAR = Linear, selection for the STT sensor type disables the cold junction compensation in the Smart Transmitter, and measurement units are in direct “mV” with the normal range from -50 to 220 mV and the extended range from -1000 to 1000 mV.
Lock: **PtBld**
Default: **Linear**
PtRes: **LMM**

Range:

PV Characterization

Range:	Valid normal range (PVEULO–PYEUHI) in Degrees C (when TCRNGOPT = Normal for SENSRTYP = Thermcpl)	Valid extended range (PVEXEULO–PVEXEUHI) in Degrees C (when TCRNGOPT = Extended for SENSRTYP = Thermcpl)
0-Linear	N/A	N/A
1-Sqrroot	N/A	N/A
2-Btherm	600 to 1650	100 to 1850
3-Etherm	-150 to 500	-2 to 1000
4-Jtherm	-100 to 750	-200 to 1200
5-Ktherm	0 to 1100	-200 to 1370
6-Rtherm	550 to 1500	0 to 1700
7-RPtherm	550 to 1500	0 to 1700
8-Stherm	550 to 1500	0 to 1700
9-Ttherm	-200 to 300	-230 to 400
10-PtDinRtd	-180 to 800	N/A
11-PtJisRtd	-180 to 650	N/A
12-NicklRtd	-45 to 315	N/A
13-CopprRtd	-20 to 250	N/A

N/A = Not Applicable

PVCLAMP

Type: **E: PVCLAMP**
Lock: **Eng/PB**
Default: **NoClamp**
PtRes: **LMM**

PVCLAMP has no effect on the PV because the LMM limits the value to a high range of 4095 and a low range of 0.

PVEUHI (AnalogIn)*Type:* **Real***Lock:* **Eng***Default:* **100.0***PtRes:* **LMM***Range:* **PVEULO to PVEXEUHI****PV High Range in Engineering Units**—Defines the PV value (in engineering units) of the high end of the PV range.**PVEULO (AnalogIn)***Type:* **Real***Lock:* **Eng***Default:* **0.0***PtRes:* **LMM***Range:* **PVEXEULO to PVEUHI****PV Low Range in Engineering Units**—Defines the PV value (in engineering units) of the low end of the PV range**PVEXEUHI (AnalogIn)***Type:* **Real***Lock:* **View***Default:* **100.0***PtRes:* **LMM***Range:* **≥ PVEUHI****PV Extended Engineering Unit Range High**—Both PVEXEUHI and PVEXEULO are used to clamp or detect a bad PV value. Refer to parameter PVEXEULO. Equals PVEUHI.**PVEXEULO (AnalogIn)***Type:* **Real***Lock:* **View***Default:* **0***PtRes:* **LMM***Range:* **PVEULO****PV Extended Engineering Unit Low Range**—The tables below show the default values in degrees C. For other engineering units, these values are appropriately converted. Equals PVEULO.

Defaults for Extended Range PV Parameters for thermocouple characterization and PVTEMP = Degrees C

PVCHAR	TCRNGOPT = NORMAL		TCRNGOPT = EXTENDED	
	PVEXEULO	PVEXEUHI	PVEXEULO	PVEXEUHI
Btherm	600	1650	100	1820
Etherm	-150	500	-200	1000
Jtherm	-100	750	-200	1200
Ktherm	0	1100	-100	1372
Rtherm	550	1500	0	1700
RPtherm	550	1500	0	1770
Stherm	550	1500	0	1700
Ttherm	-200	300	-230	400

Defaults for Extended Range PV Parameters for RTD characterization and PVTEMP = Degrees C

PVCHAR	PVEXEULO	PVEXEUHI
PtDinRTD	-180	800
PtJisRTD	-180	650
NicklRTD	-45	315
CopprRTD	-20	250

PVEXHIFL (AnalogIn)

Type: **Logical** **PV Extended High Range Violation**—Indicates that the PV has exceeded the
Lock: **View** extended-high range alarm trip point.
Default: **Off**
PtRes: **LMM**
Range: **Off** (Extended high range not exceeded)
 On (Extended high range exceeded)

PVEXLOFL (AnalogIn)

Type: **Logical** **PV Extended Low Range Violation**—Indicates that the PV has exceeded the
Lock: **View** extended-low range alarm trip point.
Default: **Off**
PtRes: **LMM**
Range: **Off** (Extended low range not exceeded)
 On (Extended low range exceeded)

PVFL (DigComp)

Type: **Logical** **PV Flag**—Indicates the current PV state as three separate Boolean parameters.
Lock: **View** PVFL(n) is On when the PV is in state "n " where n is 0, 1, or 2.
Default: **Off**
PtRes: **LMM**
Range: **Off** (PV is not in the respective state)
 On (PV is in the respective state)

PVFL (DigIn, Flag)

Type: **Logical** **PV Flag**—Represents the current PV state as a Boolean value.
Lock: **Oper**
Default: **Off**
PtRes: **LMM**
Range: **Off** [PV = STATETXT(0)]
 On [PV = STATETXT(1)]

PVFLT OPT (AnalogIn, AnalogOut, Node Specif.)

Type: **E: ?** **Enumeration of the LC Data format.**
Lock: **PtBld**
Default: **Not Configured**
PtRes: **LMM**
Range: **Signed**
 Unsigned
 Floating
 Not Configured

PVFLTOPT

PVFLTOPT (Logic)

Type: **E:PVFLTOPT** **Enumeration of the LC Data Format**— An array of (1..12) enumerations of the LC data formats.
Lock: **PtBld**
Default: **Not Configured**
PtRes: **LMM**
Range: **Boolean**
Floating
Signed
Unsigned

PVFORMAT (AnalogIn, Numeric, Timer)

Type: **E:VALFORMT** **PV Decimal Point Format**—Defines the decimal format that is to be used to display the PV value.
Lock: **Engr**
Default: **D0 for**
Timer,
D1 for
others
PtRes: **NIM**
for Numeric,
LMM for
others
Range: **0-D0 (XXXX.)**
1-D1 (XXX.X)
2-D2 (XX.XX)
3-D3 (X.XXX) (Numeric and Anal In only)

PVHIFL (AnalogIn)

Type: **Logical** **PV High Alarm Flag**—Indicates that the PV has exceeded the alarm trip point established by parameter PVHITP.
Lock: **View**
Default: **Off**
PtRes: **LMM**
Range: **Off (No PV High alarm)**
On (High PV alarm)

PVHIPR (AnalogIn)

Type: **E:ALPRIOR** **PV High Alarm Priority**—Defines the priority of the PV high alarm for this point.
Lock: **Eng/PB**
Default: **NoAction** With R670, the default value is changed to NoAction.
PtRes: **NIM**
Range: **Emergency** (Reported to all alarm summary displays)
High (Reported to Area Alarm Summary Display and Unit Alarm Summary Display)
Low (Reported to Unit Alarm Summary Display)
Journal (Logged but not reported to Universal Stations)
NoAction (Alarm is not reported to the system)

<i>Helpful Hint:</i> PVHIPR configuration requires PVHITP ≠ NaN.
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PVHITP (AnalogIn)

Type: **Real** **PV High Alarm Trip Point**—Defines the trip point for the PV high alarm for this point.
Lock: **Supr**
Default: **NaN**
PtRes: **LMM**
Range: **PVLOTP** to **PVECEUHI**, **NaN**

PVLOFL (AnalogIn)

Type: **Logical** **PV Low Alarm Flag**—Indicates that the PV has exceeded the alarm trip point established by parameter PVLOTP.
Lock: **View**
Default: **Off**
PtRes: **LMM**
Range: **Off** (PV $\geq$ Low alarm trip point)
On (PV $\leq$ Low alarm trip point)

PVLOPR (AnalogIn)

Type: **E:ALPRIOR** **PV Low Alarm Priority**—Defines the priority of the PV low alarm for this point.
Lock: **Eng/PB**
Default: **NoAction** With R670, the default value is changed to NoAction.
PtRes: **NIM**
Range: **Emergency** (Reported to all alarm summary displays)
High (Reported to Area Alarm Summary Display and Unit Alarm Summary Display)
Low (Reported to Unit Alarm Summary Display)
Journal (Logged but not reported to Universal Stations)
NoAction (Alarm is not reported to the system)

Helpful Hint: PVLOPR configuration requires PVLOTP $\neq$ NaN.

PVLOTP (AnalogIn)

Type: **Real** **PV Low Alarm Trip Point**—Defines the trip point for the PV low alarm for this point.
Lock: **Supr**
Default: **NaN**
PtRes: **LMM**
Range: **PVLLTP** to **PVEXEULO**, **NaN**

PVNORMAL (DigIn, DigComp)

Type: **E:STATETXT** **PV Normal State**—Defines the normal state of the PV using the appropriate STATETXT descriptor.
Lock: **Oper**
Default: **Off**
PtRes: **LMM**
Range: **STATETXT(0) descriptor** (Defaulted to Off for PV State 0)
STATETXT(1) descriptor (Defaulted to Off for PV State 1)
STATETXT(2) descriptor (Defaulted to Off for PV State 2; does not apply to DigIn point)

Helpful Hint: PVNORMAL change by an operator requires ALMOPT = Offnorm. PV normal state text descriptor describes the normal (desired) state, such as Run, Stop, Open, Closed.

PVNORMFL (DigIn, DigComp)

Type: Logical **PV Normal State Flag**—Indicates whether the normal state of the PV is
Lock: Oper active.
Default: Off
PtRes: LMM
Range: Off (Point is in a state other than the normal state)
On (Normal state is active)

Helpful Hint: PVNORMFL change requires ALMOPT = Offnorm. If set to On, causes text in STATETXT (1) to be used to describe the normal state of the PV, otherwise text in STATETXT (0) is used.

PVP (AnalogIn)

Type: Real **PV In Percent**—Defines the PV as a percentage of the PVEULO to
Lock: View PVEUHI range.
Default: NaN
PtRes: LMM
Range: N/A

PVRAW (AnalogIn)

Type: Real **PV Raw Value**—Indicates the raw input value of the PV before PV
Lock: View characterization is performed. The units of value for the PV are determined as
Default: NaN a percent of the raw data range (0-4095).
PtRes: LMM
Range: N/A

PVRAW (DigIn)

Type: Logical **Raw State of Field Contacts**—Indicates the current state of the field contacts,
Lock: View before input direction conversion.
Default: Off
PtRes: LMM
Range: Off
On

Helpful Hint: PVRAW is derived from the open or closed state of field contacts and from the configured direct or reverse input direction (INPTDIR).

PVRAWHI (AnalogIn)

Type: Floating **PV Raw High Value**—Indicates the high value for raw analog input data.
Lock: Engr
Default: 4095
PtRes: LMM
Range: Less than PVRAWHI + 1 (If the input data exceeds this value, it is set to NaN.)

PVRAWLO

PVRAWLO (AnalogIn)

Type: Integer **PV Raw Low Value**—Indicates the low value for raw analog input data
Lock: PtBld
Default: 0
PtRes: LMM
Range: Less than PVRAWHI - 1 (If the input data is lower than this value, it is set to NaN.)

PVSLTSRC (Timer)

Type: Integer **PV Slot Source**—Defines the address of the LC internal register for the timer
Lock: PtBld accumulator.
Default: -1
PtRes: LMM
Range: 4097 to 8191

Helpful Hint: The LC address entered in PVSLTSRC forces the address in SPSLTSRC to equal PVSLTSRC-1, therefore, PVSLTSRC and SPSLTSRC use two consecutive LC addresses.

PVSOURCE (AnalogIn, DigComp, DigIn)

Type: E:PVSOURCE **PV Source**—Defines the source of the PV input to this data point.
Lock: Oper
Default: Auto
PtRes: LMM
Range: 0-Sub (Value is provided by a sequence program)
1-Man (PV is supplied by operator or program)
2-Auto (Field wiring supplies PV)
3-Track (PV tracks OP (DigComp points only))

NOTE

If the ALMOPT parameter = Cmddis, the track option for the PVSOURCE parameter cannot be used.

Helpful Hint: PVSOURCE change by an operator requires PVSRCOPT = All

PVSRCOPT (AnalogIn, DigComp, DigIn)

Type: E:PVSRCOPT **PV Source Option**—Defines the PV source options available in this data
Lock: Eng_PB point.
Default: OnlyAuto
PtRes: LMM
Range: 0-OnlyAuto (PV source selection is not available and field wiring or memory fetch supplies PV)
1-All (PV is provided by an operator, by a sequence program, or by field wiring)

PVSTATES(0)–(4)

PVSTATES(0)–(4) (DigComp)

Type: **String_8** **PV State Descriptors**—The PV state descriptors contain the text that describes the five (0-4) possible states of a DigComp PV. The descriptors are set equal to whatever is configured in BADPVTXT and MOV PVTXT during LM node specific configuration and in STATETXT(0), STATETXT(1), and STATETXT(2) during point configuration.

Lock: **View**

Default: **None**

PtRes: **NIM**

Range: **N/A**

Helpful Hint: PVSTATES, if accessed by Control Language programs, obey the following rules:

- PVSTATES (0) = STATETXT (0)
- PVSTATES (1) = STATETXT (1)
- PVSTATES (2) = BADPVTXT
- PVSTATES (3) = MOV PVTXT
- PVSTATES (4) = STATETXT (2); does not apply unless NOSTATES = 3

PVSTS (AnalIn)

Type: **E:PVVALST** **Status of PV Input Value**—Defines the current status of the PV value.

Lock: **View**

Default: **Bad**

PtRes: **LMM**

Range: **0-Bad** (Value is bad and replaced with NaN)
1-Uncertain (Status of the value is uncertain)
2-Normal (Value is good)

PVTEMP (AnalIn)

Type: **E:TEMPTURE** **PV Temperature Scale**—Defines the temperature scale to be used in characterizing the PV input.

Lock: **PtBld**

Default: **Degrees C**

PtRes: **LMM**

Range: **Degrees C** (Celsius)
Degrees F (Fahrenheit)
Degrees R (Rankin)
Degrees K (Kelvin)

Helpful Hint: PVTEMP is to be configured when PVCHAR = TC or RTD.

PVTV (AnalIn)

Type: **Real** **PV Target Value in Engineering Units**—Defines the target value of the PV in engineering units. On Detail and Group displays on a US, this value appears as the setpoint value (SP, horizontal bar).

Lock: **Oper**

Default: **NaN**

PtRes: **LMM**

Range: **PVEXEULO** to **PVEXEUHI**,
NaN

PVTVP

PVTVP

Type: **Real**
Lock: **View**
Default: **NaN**
PtRes: **LMM**
Range: **> 0.0%, NaN**

PV Target Value in Percent—Indicates the target value of the PV as a percentage of the PVEULO to PVEUHI range.

-R-**RCASOPT (AnalogOut)**

Type: **E:\$RCASOPT** **Remote Cascade Option**—Defines whether the AM is to provide the output value for this data point. This option is available only when the point has been configured as a Full point.

Lock: **Eng/PB**

Default: **None**

PtRes: **LMM**

Range: **0-None** (No cascade mode of any type is allowed)
1-Ddc (Direct Digital Control; in cascade mode, AM point controls this point's OP)

REDTAG (AnalogOut, DigOut, DigComp)

Type: **E:REDTAG** **Red Tag State**—Allows the user to set the point as being "out of service," indicating that this point or the associated control loop needs repair or is being repaired. Once this point is put in the red tag condition, the output OP is frozen at the last value or state.

Lock: **Supr/Eng**

Default: **Off**

PtRes: **LMM**

Range: **0-Off** (Data point is in service. Point's OP is not frozen)
1-On (Data point is out of service = point's OP is frozen)

Helpful Hint: REDTAG change requires MODE = Man and MODATTR = Oper. Once a point is red tagged, parameters MODE, MODATTR, and OP (output) cannot be changed.

WARNING

The Red Tagging feature only affects the functionality of the point within the Logic Manager Module only. The PLC address (digital output, analog output) or addresses (digital composite) that a point is connected to within the PLC will NOT be frozen when the point is red tagged (REDTAG parameter is ON). For digital composite points, REDTAG being ON will NOT prevent the override interlocks from changing the OP parameter (the OP parameter represents the status within the PLC and the Red Tagging feature does not affect the PLC and its override logic). The point within the PLC must be disabled before the point is Red Tagged in the Logic Manager Module.

RINITREQ

RINITREQ (AnalogOut)

Type: **Logical** **Remote Initialization Request**—Indicates whether an initialization request
Lock: **View** has been made.
Default: **On**
PtRes: **LMM**
Range: **Off** (No request)
On (Request has been made)

<i>Helpful Hint:</i> RINITREQ does not apply if RCASOPT = None.

RV (Timer)

Type: **Real** **Remaining Time**—Indicates the amount of time remaining (in seconds or
Lock: **View** minutes) that the timer is to run.
Default: **0**
PtRes: **LMM**
Range: **0 to 65535**

<i>Helpful Hint:</i> RV represents remaining time computed as SP - PV. If SP = 0, RV is always 0.

-S-

S0BOXCLR, S1BOXCLR, S2BOXCLR (DigIn, DigComp, Flag)

Type: E:BOXCOLOR **State Box Color**—Defines the color to appear on the Group and Detail
Lock: View displays for this point when the corresponding state box is turned on.
Default: See Chart
 Below
PtRes: NIM
Range: Red
 Green
 White
 Black
 Cyan
 Yellow
 Blue
 Magenta

Point Type	Default S0BOXCLR	Default S1BOXCLR	Default S2BOXCLR
Digital Input	Yellow	Yellow	---
Digital Output	Yellow	Green	---
Digital Composite	Yellow	Green	N/A
Flag	Yellow	Green	---

SCANRATE (Node Specif.)

Type: E:\$LMMSNRT **Scan Rate**—Defines the number of times that all slots of a particular type
Lock: PtBld are scanned and processed. Refer to the *LM Control Functions* manual for
Default: AR2DT2 information on how to determine the processing capacity of the LM.
PtRes: LMM
Range:

	Analog Scan Frequency	Digital, Timer, and Logic Scan Freq. (/sec)
Null	----	----
AR1DT2	1	2
AR2DT2	2	2

CAUTION

- If you attempt to load a new SCANRATE or a new point mix from the DEB, the following items should be noted:
- if the new SCANRATE and point mix are accepted by the LMM, and if they differ in any way from their previous values, then the point database will be set to the default state and must be reloaded from the IDF.
 - If the SCANRATE or any part of the point mix is rejected by the LMM, the point mix will remain unchanged; but all points will be deleted.
 - If the SCANRATE and the point mix are equivalent to the previous values (i.e., loading a new point mix), all points in the box will be deleted.

SLOTNUM (All Point Types)

Type: **Integer** **Slot Number**—Defines the software slot number where this point resides. The processing capacity of the LM depends on the number and mix of configured control points. Refer to the *LM Control Functions* manual for a detailed description of how to determine the processing capacity of the LM, based on the amount of points being used.

Lock: **PtBld**

Default: **None**

PtRes: **NIM**

Range: **AnalogIn** (1–127 for 0.5 second scan rate, 1–254 for 1 second scan rate))
AnalogOut (1–482 for 0.5 second scan rate, 1–965 for 1 second scan rate)
DigComp (1–304)
DigIn (1–1866)
DigOut (1–4000)
Flag (LM Box Flag 1–1024)
Logic (1–14)
Numeric (LM Box Numeric 1–1024)
Timer (LM Box Timer 1–700)

SO (DigOut)

Type: **Logical**

Lock: **Oper or**
Um_Cont_Ctrl (DigOut only)

Default: **Off**

PtRes: **LMM**

Range: **Off** (Field contact is to be deenergized.)
On (Field contact is to be energized.)

SO (Timer)

Type: **Logical** **Status Output of Timer**—Indicates whether the PV (elapsed time) has reached the SP (Preset Time).

Lock: **View**

Default: **Off**

PtRes: **LMM**

Range: **On** (PV has reached the SP)
Off (PV has not reached the SP)

SO (i) (DigComp)

Type: **Logical** **Status Output Array**—Indicates the current state of the Digital Composite slot.

Lock: **View**

Default: **Off** i = 0 for state 0
i = 1 for state 1
i = 2 for state 2

PtRes: **LMM**

Range: **Off**
On

SP (Timer)

Type: **Real**
 Lock: **Oper**
 Default: **0**
 PtRes: **LMM**
 Range: **0 to 65535**

Timer Setpoint Variable—Define the preset time in seconds or minutes. If SP is 0, then PV rolls over at 65535 and continues running. The LMM monitors the LC value between write operations.

SPSLTSRC (Timer)

Type: **Integer**
 Lock: **View**
 Default: **-1**
 PtRes: **LMM**
 Range: **4096 to 8090**

Address of the LC internal register field for the Timer Preset.

Helpful Hint: The LC address entered in PVSLTSRC forces the address in SPSLTSRC to equal PVSLTSRC-1, therefore, PVSLTSRC and SPSLTSRC use two consecutive LC addresses.

ST0_OP(1)–(3) (DigComp)

Type: **Logical**
 Lock: **Eng/PB**
 Default: **Off**
 PtRes: **LMM**
 Range: **Off**
On

State 0, Outputs 1 through 3—Defines the value (On or Off) that is to be written to output number 1, 2, and 3 when the OP is in State 0.

ST1_OP(1)–(3) (DigComp)

Type: **Logical**
 Lock: **Eng/PB**
 Default: **Off**
 PtRes: **LMM**
 Range: **Off**
On

State 1, Outputs 1, 2, and 3—Defines the value (On or Off) that is to be written to output number 1, 2, and 3 when the OP is in State 1.

ST2_OP(1)–(3) (DigComp)

Type: **Logical**
 Lock: **Eng/PB**
 Default: **Off**
 PtRes: **LMM**
 Range: **Off**
On

State 2—Outputs 1, 2, and 3—Defines the value (On or Off) that is to be written to output number 1, 2, and 3 when the OP is in State 2.

STATE (Timer)**Type:** E:STATE**Lock:** View**Default:** Stopped**PtRes:** LMM**Range:** 0-Disable
1-Enable**Timer State**—Indicates the current state of the timer data point.**STATE(0)–(2) (DigComp, Flag)****Type:** String_8**Lock:** View**Default:** See Chart**PtRes:** NIM**Range:** N/A**Current State**—Indicate the current state of the data point on the displays.

These parameters represent the state text (STATETXT) descriptors as follows:

STATE0 = STATETXT(0)

STATE1 = STATETXT(1)

STATE2 = STATETXT(2); digital composite points only

Refer to the *LM Control Functions* manual for a detailed description of the digital composite point states.

Point Type	Default State0	Default State1	Default State2
Digital Input	None	None	---
Digital Output	Off	On	---
Digital Composite	None	None	None
Flag	None	None	---

STATETXT(0)–(2) (DigIn, DigComp, Flag)**Type:** String_8**Lock:** PtBld**Default:** 1 = On

0 = Off

PtRes: NIM**State Descriptor Text**—Define the states of the point using descriptors, which can have up to eight characters.

STATETXT (1) corresponds to the first ACTIVE state, or the state corresponding to PVFL = On (direct acting) or PVFL = Off (reverse acting). On the Group or Detail Display, it is in the upper box.

STATETXT (0) corresponds to the INACTIVE state, or the state corresponding to PVFL = Off (direct acting) or On (reverse acting). On the Group or Detail Display, it is the middle box for a digital composite point. For a digital input point, it is the lower box.

STATETXT (2) corresponds to the second ACTIVE state. On the Group or Detail Display for a digital composite point, it is in the lower box.

STATETXT(2) does not apply to digital input and flag points.

Range: The permissible character set for the state descriptors are as follows:

Alphabets A-Z (upper case only)

Numerics 0-9,

Underscore (_)

Helpful Hint: STATETXT has an access lock of View if PNTFORM = Componnt.
STATETXT (2) for State 2 applies only if NOSTATES = 3 for a digital composite point.

-T-**TF (AnalogIn)**

Type: **Real** **PV Filter Lag Time in Minutes**—Defines the filtering time lag to be used after
Lock: **Supr** the PV range has been checked. A value of 0.0 specifies that the PV is not
Default: **0.0 minutes** delayed.
PtRes: **LMM**
Range: **0.0 to 60.0 minutes**

-U-

UCNRECHN (Node Specif., UCN Node)

Type: **E:\$RECCHN** **UCN Receive Channel**—Indicates the channel to which the node is listening.
Lock: **View**
Default: **ChannelA**
PtRes: **LMM**
Range: **ChannelA**
 ChannelB

UNCMDL (DigComp)

Type: **Logical** **Uncommanded Change Alarm Flag**—Indicates whether an uncommanded change has been detected in the field device. (Field device has changed its state without a command.)
Lock: **View**
Default: **Off**
PtRes: **LMM**
Range: **Off** (No uncommanded change alarm)
 On (Uncommanded change alarm has been detected by this point)

UNIT

Type: **Integer** **Unit Identifier**—Defines the process unit to which this point is assigned. The unit identifier is originally assigned during network configuration, and it appears in displays and listings throughout the system.
Lock: **PtBld**
Default: **N/A**
PtRes: **NIM**

Restriction: Single-digit unit numbers are not allowed. As an example, unit 3 must be entered as 03.

Range: **A-Z, 0-9** (up to 100 unit numbers can be configured)

NOTE

A real number is returned for this parameter by the Picture Editor and by CL. This number is equivalent to the ordinal number in the enumeration list of units.

Helpful Hint: UNIT is the external parameter name. UNITNUM is the internal parameter name.

